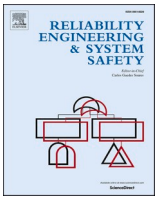
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Consideration of coupling of crack development and corrosion in assessing the reliability of reinforced concrete beams subjected to bending

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ABSTRACT

Most of the existing infrastructures (e.g., oversea bridges) are made of reinforced concrete (RC); therefore, their safety is of high priority for our society. RC structures undergo continuous deterioration due to a combination of chloride ingress and loading actions. Chloride-induced corrosion is thus one of the most serious threats. In this work a stochastic model is presented which couples degradation effects of crack development and corrosion progression based on physicochemical and mechanical models to estimate the probability of failure of a RC structure over time. To enable an efficient probabilistic assessment, a novel rapid numerical approach (RNA) is implemented for modeling the chloride diffusion. Numerical experiments show that the probability of corrosion failure is significantly underestimated if the coupled degradation effect of cracks and corrosion is not considered. Finally, parameter analyses are used to identify the effects of individual parameters, including external load level, supplementary cementitious materials and environmental temperature, on the probability of corrosion failure of RC beams over time.

1. Introduction

Chloride-induced corrosion of reinforcing bars (rebars) is one of the major threats to the safety of reinforced concrete (RC) structures (e.g., oversea bridges) exposed to corrosive environments, such as marine environments or cold environments in which de-icing salts are used [1,2]. In 2002, the annual cost of RC infrastructure corrosion was US$22.6 billion in the United States alone [3], while in 2015, the cost of anti-corrosion measures for infrastructure in Japan was 1859.2 billion yen (around US$15.5 billion in 2015) [4]. Consequently, corrosion degradation of RC is a prime concern that is addressed in concrete design codes [5,6]. For example, the Japanese Society for Civil Engineering (JSCE) uses a performance-based concrete standard to ensure that chloride-induced corrosion of RC is considered in design and construction [5]. However, due to the lack of a reliable method for assessing corrosion degradation in RC structures, most standards only provide guidance on construction measures [5,6]. Therefore, there is a need for a system that can assess corrosion-induced degradation in the performance of RC structures.

The deterioration of structural performance resulting from rebar corrosion has been extensively studied. For example, Zhong et al. [7] examined the stiffness degradation of cover concrete resulting from rebar corrosion. Lavorato et al. [8] determined the effect of rebar corrosion on the cyclic behavior of RC structures. Pelle et al. [9] explored the time-dependent deterioration of RC bridge columns resulting from the combined effects of chloride ingress and mechanical performance degradation. However, the process of deterioration of RC structures involves great uncertainty and randomness due to variations in material properties (e.g., concrete strength), structure geometry (e.g., cover depth), and the highly variable nature of the marine environment (e.g., surface chloride content) [10,11]. Therefore, a probabilistic approach must be applied for evaluating the performance of corroded RC structures.

Several probabilistic models have been developed to explore degradation of the mechanical performance of RC structures resulting from rebar corrosion. For example, Stewart [12] studied the effects of spatial variability of rebar pitting corrosion the reliability of RC beams in

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List of Notations

b_c	Width of concrete beam (mm)
b_{cl}	Intermediate variable indicating the free chloride content
d_c	Concrete cover depth (mm)
d_{Re}	Diameter of rebar (mm)
f_{O_2}	Correction factor for the Tafel slope of the reduction reaction
f_y	Yield strength of non-corroded rebar (MPa)
$f_c^{t_0}$	Compressive strength of concrete at reference age t_0 (MPa)
$f_c^{t_i}$	Compressive strength of concrete (MPa)
f_c^u	Ultimate compressive strength of concrete (MPa)
$f_{y,cor}^{t_i}$	Yield strength of corroded rebar (MPa)
h_0	Effective height of the cross section of concrete beam (mm)
h_c	Height of concrete beam (mm)
$h_{cr}^{t_i}$	Thickness of cracked concrete layer (mm)
$h_{con}^{t_i}$	Thickness of uncracked concrete layer (mm)
m_{cr}	Adjustment factor for the chloride diffusion coefficient of cracked concrete layer
m_{cor}^T	Adjustment factor for the temperature-dependence of rebar corrosion
$m_{T_0}^{T_a}$	Adjustment factor for the temperature-dependence of chloride diffusion coefficient
$m_{x=0}^C$	Adjustment factor for the surface chloride content of concrete
$p_{cor}^{t_i}$	Corrosion rate of rebar (mm/day)
pH^{t_i}	pH of the pore solution of cement paste
pH^{t_0}	pH of the pore solution of cement paste at reference age t_0
pH^u	Ultimate pH of pore solution
t_i	Time at which the ith step occurs (day)
t_0	Reference age (day)
t_b	Time variable used in the calculation of the surface chloride content (day)
t_{st}	Initial exposure time (day)
t_{ex}	Exposure time (day)
t_{Hy}	Time variable used in the calculation of pH and compressive strength (day)
t_{Hy}^u	Time to hydration completion (day)
x	Interested concrete depth in the analytical solution (mm)
x_s	Distance from the neutral axis of a concrete beam's cross section to its rebar (mm)
A_n	Parameter considering the influence of existing chloride content
A_{Re}^c	Cross-section area of a rebar (mm^2)
A_0^s	Surface area of the reference rebar (mm^2)
A_{Re}^s	Surface area of a rebar
$A_{cor}^{t_i}$	Pitting corrosion area of a rebar (mm^2)
$A_{cor}^{D_{Re}^D}$	Pitting corrosion area when the divided number is D_{Re}^D.
A_{cor}^{250}	Pitting corrosion area when D_{Re}^D is 250
$\overline{C}(x)$	Chloride profile in the last step
C_{cement}	Free chloride content of raw cement and supplementary cementitious materials (wt% binder)
C_{Fe}	Concentration of dissolved iron (μmol/L)
C_{O_2}	Concentration of dissolved oxygen (μmol/L)
$C_{x=0}^a$	Initial surface chloride content of concrete (wt% binder)
$C_{x=0}^b$	Characteristic surface chloride content of concrete indicating time-dependence (wt% binder)
$C_{x=0}^{t_i}$	Surface chloride content of concrete (wt% binder)
C_{cor}^{free}	Totally free chloride content at rebar (wt% binder)
C_{Re}^{free}	Free chloride content at rebar due to chloride ingress (wt% binder)
C_{Re}^{b,t_i}	Bound chloride content of concrete at rebar (wt% binder)
D_1	Chloride diffusion coefficient of the first layer
D_2	Chloride diffusion coefficient of the second layer
D_{Re}^D	Divided number for the rebar diameter for determining pitting corrosion area
$D_{cr,k}$	Characteristic chloride diffusion coefficient of cracks in concrete (mm^2/day)
$D_{cr}^{t_i}$	Chloride diffusion coefficient of the cracked concrete layer (mm^2/day)
$D_{con}^{t_i}$	Chloride diffusion coefficient of the uncracked concrete layer (mm^2/day)
E_{cor}	Thermal activation energy of rebar corrosion (J/mol)
E_s	Elastic modulus of rebar (GPa)
E_D	Thermal activation energy of chloride diffusion (J/mol)
E_{Fe}	Corrosion potential of the iron oxidation reaction (V)
E_{Fe}^{Θ}	Reference corrosion potential of the iron oxidation reaction (V)
E_{O_2}	Corrosion potential of the oxygen reduction reaction (V)
$E_{O_2}^{\Theta}$	Reference corrosion potential of the oxygen reduction reaction (V)
Err_{cor}	Relative error for D_{Re}^D
F	Faraday's constant (9.65×10^4 C/mol)
F_I^m	Transformation of corrosion current (A/m^2) to corrosion mass rate (mm/day)
$I_{cor}^{t_i}$	Corrosion current of rebar (A/m^2)
I_{Fe}^{Θ}	Reference corrosion current at the anode (1.0×10^{-5} A/m^2), which is used to adjust the corrosion potential
$I_{O_2}^{\Theta}$	Reference corrosion current at the cathode (1.0×10^{-10} A/m^2), which is used to adjust the corrosion potential
N	Number of characteristic roots for a rapid numerical approach (RNA)
M	Total steps for simulation
M_{cor}	Number of corroded rectangles used to determine pitting corrosion area
M_T	Number of total rectangles used to determine pitting corrosion area
P_{Cl}	Adjustment factor for the corrosion potential of the iron oxidation reaction, to account for the influence of free chloride in pore solution
P_d^I	Distance from the focal rectangle (x_I, y_I) to the bottom rectangle (x_B, y_B)
$P_{cor,avg}^{t_i}$	Average corrosion depth of rebar (mm)
$P_{cor,pit}^{t_i}$	Pitting corrosion depth (mm)
R	Molar gas constant (8.314 J/K/mol)
R_{pit}	Ratio of pitting corrosion depth to average corrosion depth
RH	Environmental humidity (%)
$S_p^{t_i}$	Saturation of concrete at rebar
T_a	Absolute temperature of the environment (K)
T_{En}	Environment temperature (°C)
T_0	Reference temperature 20 °C
V/S	Ratio of volume to surface area of a concrete beam (mm)
W	Water amount for casting concrete (kg/m^3)
X^{t_0}	The superscript t_0 represents a variable at reference time t_0
X^{t_i}	The superscript t_i represents a variable at time t_i
Z_{Fe}	Valence electron of iron atom
Z_{O_2}	Valence electron of oxygen atom
α_{cr}	Time-dependence of the chloride diffusion coefficient of cracked concrete
α_{con}	Time-dependence of the chloride diffusion coefficient of uncracked concrete
α_{pit}	Standard variation of the Gumbel distribution of reference corroded rebar
α_y	Coefficient of the influence of corrosion on yield strength
α_{cl}^b	Coefficient of chloride binding capacity of cement paste

α_{Hy}	Hydration rate factor
β	Correction for compression height when the compressive range is approximately rectangular
γ'	Correction for compressive stress when the compressive range is approximately rectangular
γ_{EM}	Euler–Mascheroni constant
Δt	Interval time for RNA (year)
Δx	Interval depth of the chloride profile for RNA (mm)
$\varepsilon_c^{t_i}$	Concrete strain
$\varepsilon_s^{t_i}$	Rebar strain
$\varepsilon_{sh}^{t_i}$	Drying shrinkage strain of concrete
ε_{sh}^{u}	Ultimate drying shrinkage of concrete
η_{Fe}	Tafel slope of the reduction reaction
η_{O_2}	Tafel slope of the oxidation reaction
λ_n	Characteristic roots of RNA
μ_0	Mean value of the Gumbel distribution of reference corroded rebar
$\mu_{R_{pit}}^{Gum}$	Mean value of the Gumbel distribution
μ_{Pit}^{M}	Correction for the influence of the surface area of rebar
σ_s	Rebar stress (MPa)
$\sigma_{R_{pit}}^{Gum}$	Standard variation of the Gumbel distribution
$\phi_{con}^{t_i}$	Concrete porosity

flexure. Dizaj et al. [13] devised a fiber-based nonlinear finite-element modeling system to study the effect of rebar corrosion on the hysteretic behavior of RC frames. However, the application scope of these models is limited because they do not consider the effect of chloride ingress into concrete.

To address the limitations of the above-mentioned models, many probabilistic approaches have been developed to study the deterioration processes of RC structures coupled with chloride ingress. For example, there have been extensive investigations of the fragility of RC bridges subjected to combined deteriorative effects of rebar corrosion and earthquakes [7,14–22] . Choe et al. [21,22] have devised a formulation for estimating the seismic fragility of corroded RC bridges. Similarly, Zhong et al. [7,14] have developed a method to estimate the seismic fragility of corroded RC bridges that accounts for both the loss of reinforcing steel and the stiffness degradation of cover concrete. Kumar et al. [16] constructed a stochastic framework based on semi-analytic solutions to simulate the process of deterioration of RC bridges. This framework considers both the gradual deterioration process of rebar corrosion and the shock damage of earthquakes. Jia et al. [15] subsequently extended this stochastic framework to account for changes in the state of variables of RC bridges (such as material properties, member dimensions, and imposed boundary conditions) due to various deterioration processes. Xu et al. [17,18] have devised a random field-based analytical system that simultaneously accounts for the inhomogeneous spatial variability, intra-level correlation, inter-level dependency, and temporal variability of RC bridges. Reliability analyses have also been used to estimate the life-cycle cost and maintenance cost of RC bridges [19,20]. In other work, Chen et al. [23] developed an evidence-based analytical model to predict the time of initiation of corrosion due to chloride ingress. Francesco et al. [24] examined the reliability of a nuclear power plant subjected to RC deterioration caused by chloride ingress, while Hackl et al. [25] developed a system for assessing RC reliability that considers the coupled effect of corrosion initiation and progression on RC. However, as all of these studies conducted probabilistic analysis based on the chloride ingress of uncracked concrete, they have underestimated deterioration due to the coupling of chloride ingress and loading actions and have thus overestimated the structural reliability of RC that is exposed to chloride ingress [26,27].

RC structures undergo continuous deterioration due to a combination of chloride ingress and loading actions [1,28,29], as was confirmed by a 12-year observational study of the condition of RC beams subjected to flexural load in a confined salt fog [29]. This deterioration involves RC cracking under external loads, with these cracks providing preferential penetration paths for chloride ingress [29,30] that results in pitting corrosion of rebars. This corrosion increases the stress on rebars, thereby increasing the width and depth of cracks, which facilitates continued chloride ingress and leads to increased corrosion of rebars [29]. Thus, the coupled effects of crack development and corrosion progression continuously degrade the mechanical capacity of RC structures and thus must be considered when assessing the whole-life performance of such structures [29].

Several probabilistic approaches have been developed to assess the coupled effects of crack development and rebar corrosion on the reliability of RC structures. For example, Kwon et al. [26] studied the effect of early-age cracks induced by hydration heat and shrinkage on the service life of RC structures. Vu et al. [27] developed a two-dimensional spatiotemporally dependent reliability model to predict the probability and extent of corrosion-induced cracking of an RC surface. Arteaga et al. [31] devised a reliability model that considers the coupled effects of chloride ingress and RC cracking on RC structures. However, due to the lack of appropriate mathematical methods to describe the coupled effect of crack development and corrosion progression on RC structures, these studies have used empirical models to approximate the processes of deterioration of RC structures. This limits the application scope and reliability of the above-mentioned approaches and models. Therefore, a more general analytical approach must be developed to simulate the coupled effects of these two processes on the whole-life performance of RC structures.

The finite element method (FEM) is a generic way to simulate the coupled degradation process of crack development and corrosion progression. However, the FEM is time-consuming, and probabilistic analyses require millions of samples (e.g., Monte Carlo method) to achieve convergent results, which makes the FEM inapplicable due to the huge computational cost [11]. Therefore, a computationally efficient approach is required for probabilistic analyses.

Furthermore, supplementary cementing materials (SCMs), such as fly ash, slag and calcined clay with limestone, have also been widely used to improve the durability of RC structures subjected to chloride ingress [32–34]. Limestone calcined clay cement (LC3) is a recently developed SCM that is popular in the construction industry due to its low emissions of carbon dioxide. SCMs have complex effects on rebar corrosion [32–37]; in particular, as SCMs react with calcium hydroxide to produce an aluminum-modified calcium silicate hydrate (C-(A)-S-H) gel, the microstructure of SCM-blended cement paste is refined and its chloride diffusion coefficient decreases [32,33]. The chloride binding capacity of SCMs blended cement paste is also enhanced due to the incorporation of Al elements [34]. Moreover the pH of pore solution decreases as a result of the pozzolanic reaction of SCMs [35], which increases the vulnerability of the passive layer of rebars to chloride ingress [36,38]. However, most probabilistic models cannot consider SCM-blended concrete because they are based on empirical rebar corrosion models, which do not consider the effects of the hydroxide-to-chloride ratio of pore solutions and the chloride binding capacity of cement paste on RC structures. Therefore, a comprehensive system for the reliability assessment of RC must include a rebar corrosion model that accounts for the effects of these two factors on RC structures.

Advances in mathematical modeling over the past decade have enabled the development of many numerical techniques. For example, a rapid numerical approach (RNA) was developed by Wang et al. [11] to

simulate the time-dependent process of chloride diffusion through two layers of materials (e.g. surface-treated concrete). This RNA required minimal computation effort to simulate chloride ingress into cracked concrete [39]. In addition, Hussain et al. [40] developed a corrosion rate model considering hydroxide-to-chloride ratio, while Ishida et al. [28, 41] have developed a model of the chloride binding of cement paste. Moreover, Stewart [12] developed a pitting corrosion model. All these models can be used to simulate corrosion initiation and progression.

This study validates the above-mentioned chloride diffusion and corrosion rate models by comparing their predictions with the results of finite element method (FEM) modeling and experimental studies, respectively. In addition, this study devises an efficient and comprehensive reliability analysis system based on these advanced models. This system can consider various types of SCM-blended concrete and the coupled degradation effects of crack development and corrosion progression. As such, this system can be used to assess the whole life performance of RC structures without the need for empirical deterioration models. Finally, the study uses parameter analyses to investigate the effect of different parameters – external load level, SCMs and environmental temperature – on the corrosion failure probability (CFP) of RC beams. In summary, this reliability analysis system provides reasonable guidelines for civil engineers seeking to optimize the design and maintenance schedules of RC structures.

2. Reliability analysis of RC structures and the coupled degradation effect

RC structures undergo continuous deterioration due to a combination of chloride ingress and loading actions. This study develops a reliability assessment system that estimates the failure time of RC structures under the coupled degradation effects of crack development and corrosion progression (Fig. 1).

The required inputs are the water-binder ratio, temperature, the rebar properties, concrete strength, pH of a pore solution, geometry of a RC structure and the boundary conditions to which the RC structure will be exposed during its life cycle. These parameters are assigned probabilistic distributions in Monte Carlo simulations (Fig. 1a), in each of which the chloride diffusion profile of cracked concrete is first represented as a rapid two-layer diffusion model (Fig. 1b). Then, based on the chloride content of rebar, the corrosion rate and pitting corrosion area are determined using corrosion rate and pitting corrosion propagation models (Fig. 1c). Next, the widths and thicknesses of the cracks are determined through mechanical equilibrium analyses of the corroded rebar (Fig. 1d), and these crack geometry data are used as the input for the next step in the estimation of chloride ingress. The above steps are repeated until the stress of the rebar is higher than its yield strength (Fig. 1e), i.e., until it fails, and the time to failure is recorded. Finally, the results of the Monte Carlo simulation enable the CFP of the RC to be determined (Fig. 1f). Because this system is developed based on physicochemical and mechanical models, the whole-life performance of RC structures can be comprehensively assessed without the empirical deterioration models for concrete. The details of each model are explained in Sections 2 to 5.

3. Numerical model of chloride diffusion in cracked concrete

3.1. Rapid numerical approach of chloride diffusion in cracked concrete

Cracked concrete can be modelled as a two-layered material consisting of a cracked concrete layer and an uncracked concrete layer, as shown in Fig. 2. The thickness and the chloride diffusion coefficient of cracked concrete are denoted as $h_{cr}^{t_i}$ and $D_{cr}^{t_i}$, whereas the thickness and the chloride

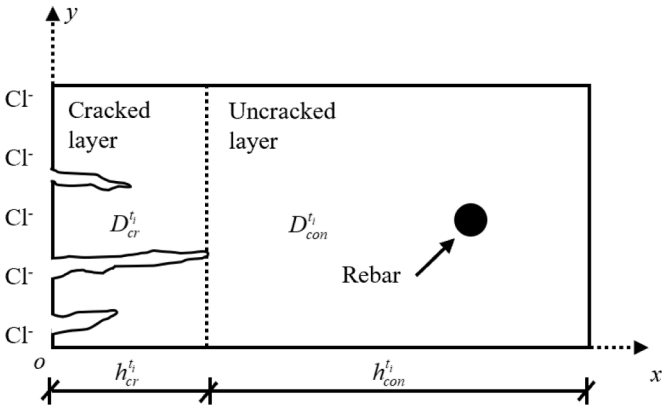


Fig. 2. Chloride (Cl^-) diffusion model for cracked RC.

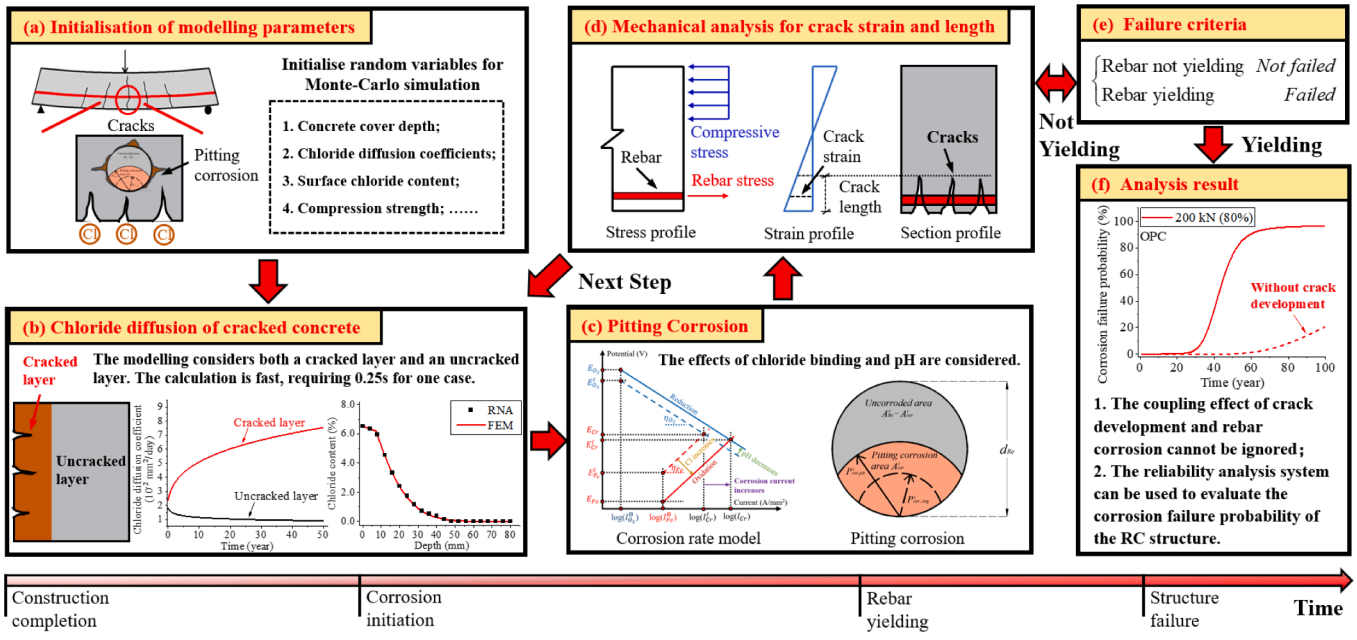


Fig. 1. Reliability assessment system for RC structures subjected to chloride ingress. *Note*: Symbol definitions are given in the list of notations.

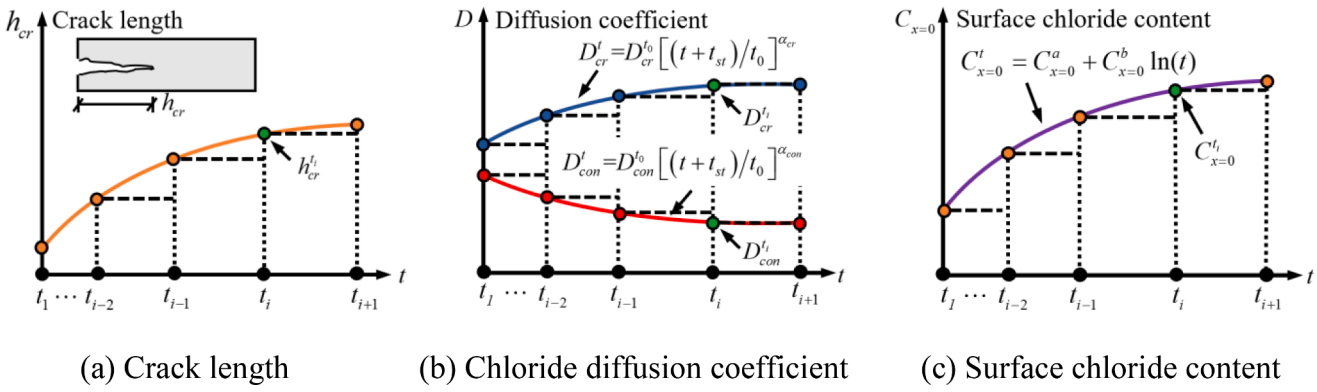


Fig. 3. Numerical simulations of time-dependent properties of cracked RC.

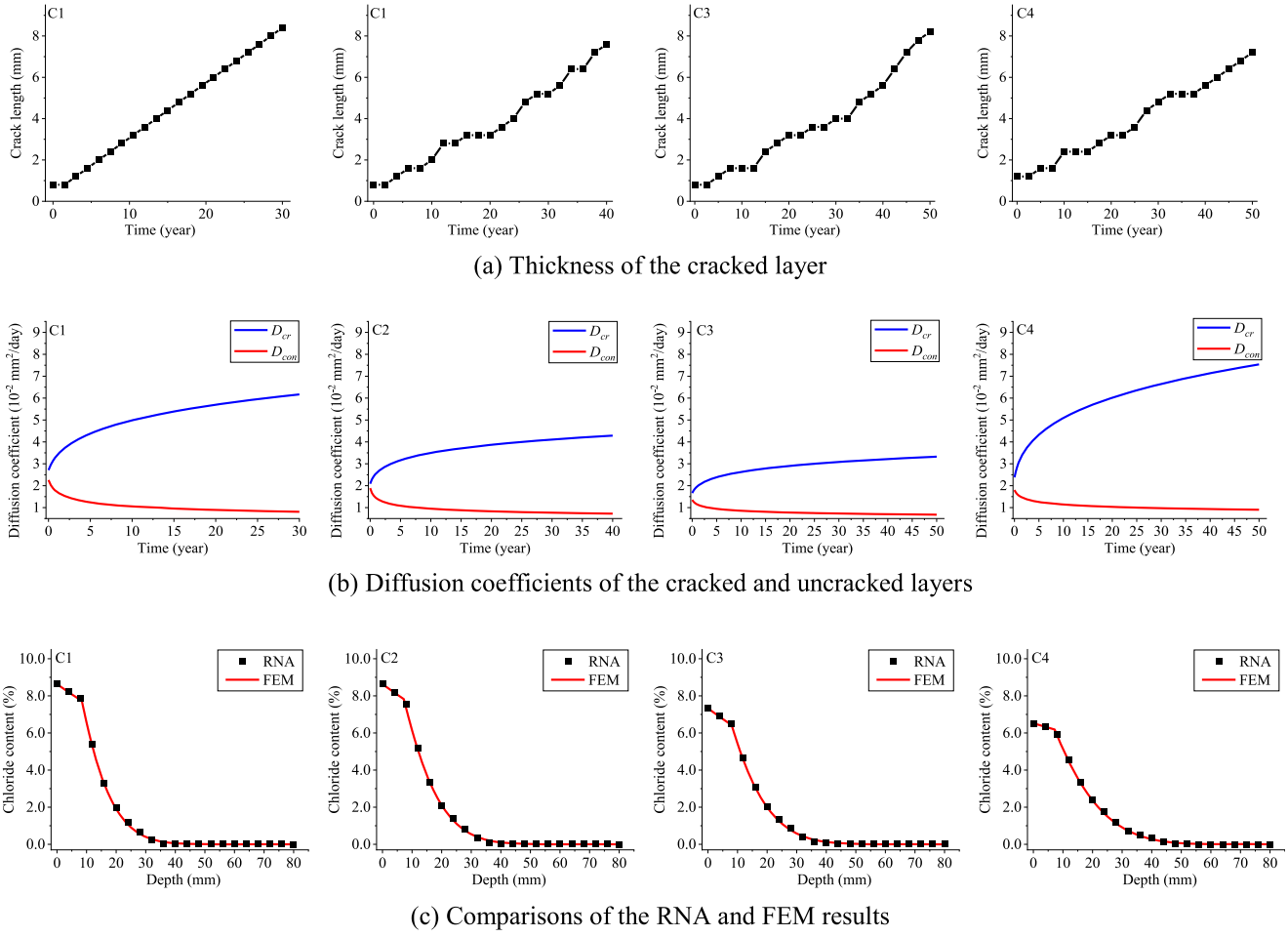


Fig. 4. Validation of RNA for estimating chloride diffusion in cracked RC.

diffusion coefficient of uncracked concrete are denoted as $h_{con}^{t_i}$ and $D_{con}^{t_i}$. t_i and t_0 denote the time at the ith step and reference time, respectively.

Concrete cracks under an external load, and these cracks provide preferential paths for chloride ingress [30]. Over time, chloride accumulates on the surface of RC, but the porosity and chloride diffusion coefficient of the uncracked layer decrease as the cement hydration process proceeds. To estimate the diffusion of chloride in cracked concrete, it is necessary to consider crack geometries and the time-dependent processes of chloride diffusion coefficient and surface chloride content.

An explicit analytical solution was developed for simulating the diffusion process in two-layered materials (Appendix A) [11,42]. However, this solution is only valid for two-layered materials that have a constant chloride diffusion coefficient in each layer (e.g., short-term salt-water ponding tests of surface treated concrete) [11,42]. Wang et al. [11] subsequently proposed a rapid numerical approach (RNA) in order to extend the analytical solution to simulate time-dependent chloride diffusion process in two-layered materials: all time-dependent parameters (e.g. chloride diffusion and surface chloride content) are approximated using constant step functions. Because the chloride diffusion coefficient is approximated as a constant in each time step, the analytical solution can be used. The numerical solution thus better approaches the correct result as the number of time steps increases. This

Table 1
Parameters used in the validation of the chloride diffusion of cracked concrete.

Case	$D_{cr}^{t_0}$ (mm^2/day)	$D_{con}^{t_0}$ (mm^2/day)	α_{cr}	α_{con}	$C_{x=0}^{a}$ (wt% binder)	$C_{x=0}^{b}$ (wt% binder)	t_{ex} (year)	t_{st} (day)	t_0 (day)
C1	2.5×10^{-2}	2.5×10^{-2}	0.20	−0.25	3.0	3.5	30.0	180	120
C2	2.0×10^{-2}	2.0×10^{-2}	0.15	−0.20	3.0	3.5	40.0	120	90
C3	1.5×10^{-2}	1.5×10^{-2}	0.15	−0.15	2.5	3.0	50.0	180	90
C4	2.0×10^{-2}	2.0×10^{-2}	0.25	−0.15	2.5	2.5	50.0	180	90

Table 2
Computational times in the finite element method (FEM) and rapid numerical approach (RNA).

	C1	C2	C3	C4	Average
FEM (s)	5,332	6,589	7,680	7,577	
RNA (s)	0.2048	0.2301	0.3008	0.2642	
FEM/RNA	26,035	28,635	25,532	28,679	27,220

approach was used to successfully estimate the time-dependent chloride diffusion process of surface-treated concrete, which comprises a surface-treated layer and a concrete substrate [11].

RNA can also be used to simulate the time-dependent chloride diffusion process in cracked concrete [39]. The time-dependent parameters of crack length, diffusion coefficient and surface chloride content are approximated using the constant step functions, as shown in Fig. 3. Four case studies with different parameters were conducted to validate the use of RNA for predicting chloride ingress into cracked concrete. The crack development is described using 10-point step functions Fig. 4a); the chloride diffusion coefficients of the cracked and uncracked concrete are described using power functions (Eqs. (1) and ((2), Fig. 4b); and, following Chalee et al. [43], the surface chloride content is described using a logarithmic function (Eq. (3)). It should be noted that Eq. (1) is only used for validation. A more reasonable model for the chloride diffusion coefficient of cracked concrete is introduced in Section 5.1. Because the logarithmic function for the surface chloride content (Eq. (3)) is based on a 5-year field investigation, the value of the exposure time for surface chloride content t_b is limited to prevent the underestimation or overestimation of surface chloride content.

$$D_{cr}^{t_i} = D_{cr}^{t_0}[(t_i + t_{st})/t_0]^{\alpha_{cr}} \quad i = 1, 2, 3.....M \tag{1}$$

$$D_{con}^{t_i} = D_{con}^{t_0}[(t_i + t_{st})/t_0]^{\alpha_{con}} \quad i = 1, 2, 3.....M \tag{2}$$

$$C_{x=0}^{t_i} = C_{x=0}^{a} + C_{x=0}^{b}\ln(t_b)\ , \begin{cases} t_b = 180 & t_i \le 180 \\ t_b = t_i & 180 < t_i < 1825 \\ t_b = 1825 & t_i \ge 1825 \end{cases} \quad i = 1, 2, 3.....M \tag{3}$$

where M is total steps for simulation; $D_{cr}^{t_i}$ and $D_{con}^{t_i}$ are the diffusion coefficients of cracked and uncracked concrete;$D_{cr}^{t_0}$ and $D_{con}^{t_0}$ are the diffusion coefficients of cracked and uncracked concrete, respectively, at reference age t_0; $C_{x=0}^{a}$ is the initial surface chloride content of the concrete (wt% binder); $C_{x=0}^{b}$ is the characteristic surface chloride content of concrete indicating time-dependence (wt% binder); α_{cr} and α_{con} are the time-dependent chloride diffusion coefficients of the cracked and uncracked concrete, respectively; t_{st} is the start time of the exposure of the concrete structure to a chloride environment; and t_i is the duration of the exposure of the RC structure to a chloride environment.

3.2. Validation of the RNA for analysing chloride diffusion in cracked concrete

The RNA is validated by comparing the results with those of the FEM developed by well-established procedures [44]. Table 1 summarises the values of the parameters used in the validation. The FEM mesh is one-dimensional, where the element length and the time step are set as 0.4 mm and 0.1 day [11], respectively. The computational accuracy of the RNA increases as: i) the number of characteristic roots (N) increase; ii) the time interval (Δt) decreases; and iii) the depth interval of chloride profile (Δx) decreases [11]. Following other convergent analyses [11], $N = 10$, $\Delta t = 1$ year and $\Delta x = 1$ mm have been used for validation. All the analyses are conducted on a MATLAB platform.

The numerical results as shown in Fig. 4c demonstrate that there is a good agreement between the two estimates, confirming the validity of the RNA. Table 2 compares the computational times between the RNA and the FEM (CPU Intel i5–8265 U, 1.60 GHz), which shows that the RNA is more than 27,000 times faster than the FEM. This highly efficient approach enables the reliability analysis based on Monte Carlo simulations.

4. Pitting corrosion model of rebar

4.1. Corrosion rate model of rebar

Cement paste is an alkaline material, as the pH of pore solution ranges from 12.5 to 13.7 [35]. This high alkalinity of the pore solution leads to the formation of a passive layer (e.g., a ferric oxide film) on a rebar surface, which provides protection against corrosion. However, over time, free chloride ions in the pore solution destroy the passive layer, causing the rebar to corrode [35]. The pH of the pore solutions of SCM cement paste is lower than that of ordinary Portland cement (OPC) paste due to the pozzolanic reaction. As a lower pH makes the passive layer of rebar more vulnerable to chloride ions, the corrosion rate of rebar increases as the chloride content increases and the pH decreases [36,38]. Therefore, corrosion rate models should consider the effects of both pH and chloride content.

The corrosion rate model of Hussain et al. [28,40], based on an Evans' diagram (Fig. B1), assesses the influence of pH and chloride content of cement paste by adjusting the potential reduction and oxidation reactions (see Appendix B). This study adopts this model.

$$p_{cor}^{t_i} = F_I^m \times S_p^{t_i} \times \phi_{con}^{t_i} \times I_{cor}^{t_i}\left(C_{\mathrm{Re}}^{free,t_i}\right) \times m_{cor}^{T} \tag{4}$$

$$m_{cor}^{T} = \exp\left[\frac{E_{cor}}{R} \times \left(\frac{1}{273.15 + T_0} - \frac{1}{T_a^{t_i}}\right)\right], \quad \phi_{con}^{t_i} = \frac{0.284}{\exp\left(f_c^{t_i}/37.1\right)} \tag{5}$$

where $p_{cor}^{t_i}$ is the corrosion rate of rebar (mm/day); F_I^m is the transformation of the corrosion current (A/m^2) to the corrosion mass rate (mm/day) for pitting corrosion model, which is 3.2×10^{-3}; $S_p^{t_i}$ is the saturation degree of the concrete at the rebar; $I_{cor}^{t_i}$ is the corrosion current of the rebar (A/m^2), which is determined by Eq. (B1); m_{cor}^{T} is the temperature-dependent aspect of rebar corrosion determined by the Arrhenius equation [28,40]; E_{cor} is the thermal activation energy of the corrosion reaction, which is assumed to be 9×10^4 J/mol; R is the molar gas constant (8.314 J/K/mol); and T_a is the absolute temperature of the environment, which is $T_a = 237.15 + T_{En}$ (K), where T_{En} is the temperature of the environment (°C) and T_0 is the reference temperature (20 °C).

Aggressive agents penetrate concrete more easily when the concrete has a high porosity, and this is accounted for in Eq. (4). It is well known that the compressive strength of concrete increases as its porosity decreases [45,46]. Schiller [45] derived a logarithmic formula to describe the relationship between compressive strength and concrete porosity. Kumar et al. [46] recently verified this formula by comparing the in-situ compressive strengths of concrete specimens, and then developed an empirical model (Eq. (5)). For simplicity, this study determines concrete

porosity ($\phi_{con}^{t_i}$) using this compressive strength model.

4.2. Validation of the corrosion rate model of rebar

The corrosion rate model is validated by comparing its performance with the results of corrosion tests of RC specimens conducted by Hussain et al. [28,40]. In that study, rebars with a diameter of 13 mm were embedded in the middle of the concrete specimens. The RC specimens were mixed with sodium chloride (0–6% of binder) at the beginning. To investigate temperature effect, RC specimens were placed in environment chambers at different temperatures (20 °C, 40 °C and 60 °C). After 200 days, the specimens were split and the corrosion degree of the rebar was measured using the gravimetric method. More details have been given in [28,40]. The concrete porosity is assumed to be 0.1034 at 28 days, based on Eq. (5) (i.e., $f_c = 37.5$MPa).The saturation is assumed to be 0.80, based on water transport analyses using a thermal dynamic multi-scale water transport model [47]. The corrosion rates of the experimental specimens and the model predictions are summarised in Fig. 5. It demonstrates a satisfactory agreement, where the maximum error is less than 18% in common conditions (i.e., G_{Re}^{free} and T_{En} are in the ranges of (0.25%, 3.3%) and (20 °C, 40 °C), respectively). This model can thus be used to accurately estimate the corrosion rate of rebars.

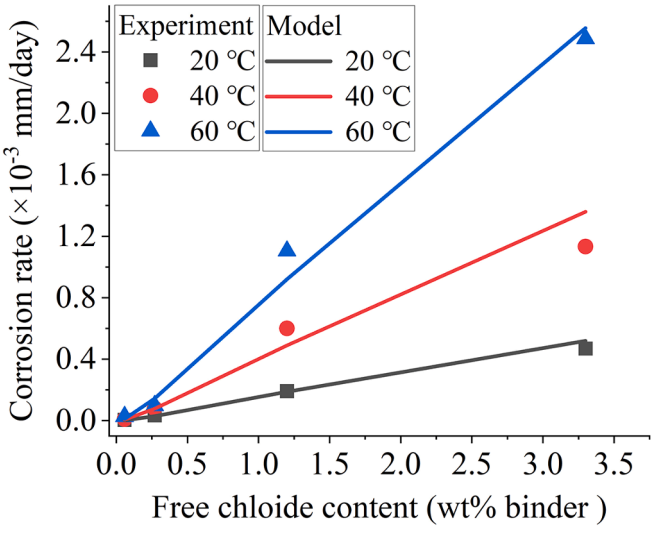


Fig. 5. Comparison of experimental and model corrosion rates.

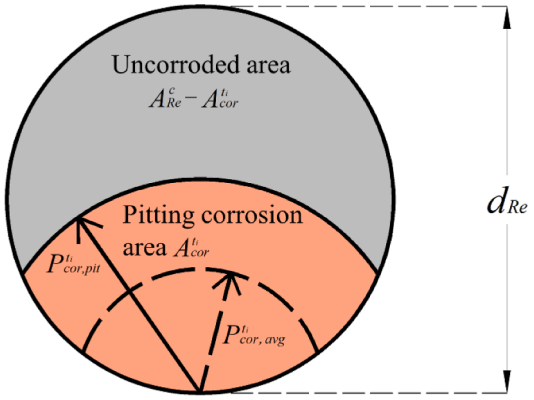


Fig. 6. Pitting corrosion of rebar.

4.3. Propagation of pitting corrosion

Based on the corrosion rate model given in Eq. (4), the average corrosion depth $P_{cor,avg}^{i}$ (Fig. 6) can be determined as follows:

$$P_{cor,avg}^{t_i} = P_{cor,avg}^{t_{i-1}} + p_{cor}^{t_i}\Delta t \quad i = 1, 2, 3.....M \tag{6}$$

However, chloride-induced corrosion is a typical type of pitting corrosion (localised corrosion), and thus the maximum corrosion depth $P_{cor,pit}^{t_i}$ exceeds the average corrosion depth $P_{cor,avg}^{t_i}$ [12]. Gonzalez et al. [48] found that the ratio R_{pit} Eq. (7) varied from 4 to 8. Stewart [12] suggested that the ratio R_{pit} follows a Gumbel distribution, where the Gumbel statistical parameters can be determined using Eqs. (8) and (9).

$$P_{cor,pit}^{t_i} = R_{pit} P_{cor,avg}^{t_i} \tag{7}$$

$$\mu_{R_{pit}}^{Gum} = \mu_{R_{pit}}^{M} + \gamma_{EM}\alpha_{pit} \;\; ; \;\; \sigma_{R_{pit}}^{Gum} = \frac{\pi}{6}\alpha_{pit} \tag{8}$$

$$\mu_{Pit}^{M} = \mu_0 + \frac{1}{\alpha_{pit}} \ln\left(\frac{A_{Re}^{s}}{A_0^{s}}\right) \tag{9}$$

where $\mu_{R_{pit}}^{Gum}$ is the mean value of the Gumbel distribution; $\sigma_{R_{pit}}^{Gum}$ is the standard variation of the Gumbel distribution; γ_{EM} is the Euler–Mascheroni constant, which can be taken as 0.5772; μ_{Pit}^{M} is a correction factor that considers the effects of the surface area of rebar A_{Re}^{s}; μ_0 and α_{pit} are the mean and standard variation of the Gumbel distribution in the reference corroded rebar; and A_0^{s} is the surface area of the reference rebar. μ_0 and α_{pit} are set as 5.08 and 1.02, respectively, when A_0^{s} is the surface area of an 8-mm-diameter rebar with a length of 125 mm (i.e., 3,140 mm^2) [12].

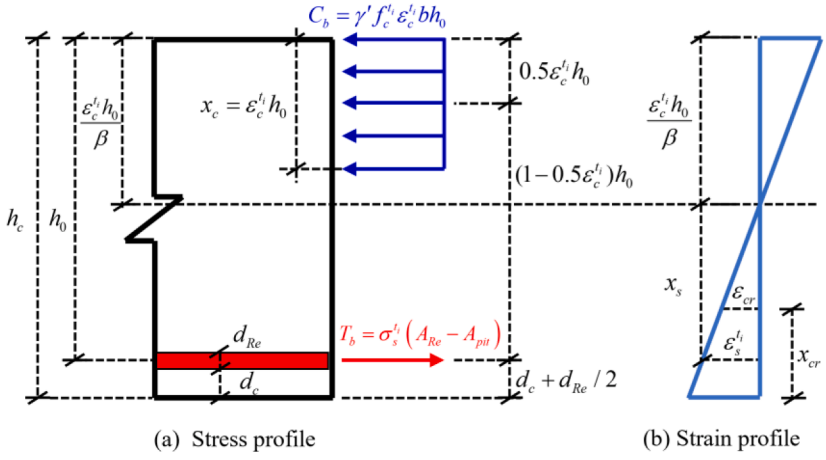


Fig. 7. Stress and strain profile of a rebar cross section.

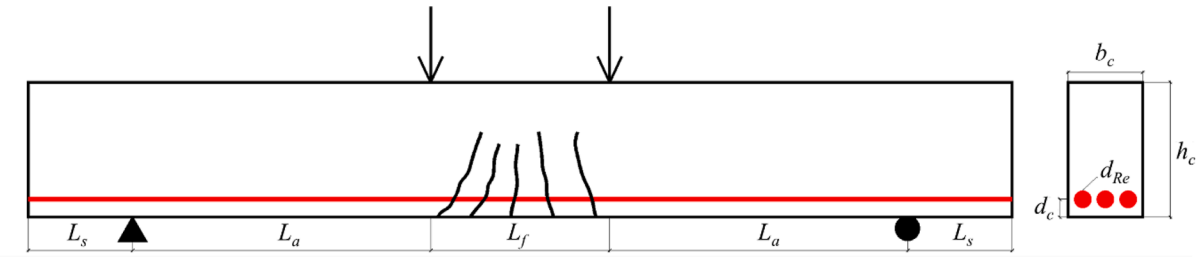


Fig. 8. Reinforced concrete beam with a flexural load.

Therefore, the pitting corrosion area ($A_{cor}^{t_i}$) of rebar with a diameter of d_{Re} can be obtained, as shown in Fig. 6. An analytical solution was developed by Stewart [12] to determine $A_{cor}^{t_i}$. However, as this solution requires complicated calculations, a more efficient numerical approach was developed for determining the pitting corrosion area by meshing method (see Appendix C).

5. Flexural analysis of corroded RC beams

5.1. Mechanical equilibrium conditions under flexural moment

As flexure is the predominant cause for failure and safety margin in shear is higher based on the reliability analysis of an existing RC bridge [1,49], this study simplifies the reliability analysis system by considering an RC beam subjected to a flexural moment. Fig. 7 shows the stress and strain profile of the cross section of an RC beam subjected to primarily bending moment. For simplicity, the compressive range has been considered as approximately rectangular with a height of x_c [50]. Given the force equilibrium conditions Fig. 7a, the stress σ_s on a rebar can be determined by Eqs. (10) ~ (12):

$$\begin{cases} M_f = \gamma' f_c^{t_i} b_c x_c (h_0 - 0.5x_c) \\ \sigma_s \left(A_{Re}^c - A_{cr}^{t_i}\right) = f_c^{t_i} b_c x_c \\ \varepsilon_c^{t_i} = \sigma_s \rho_{s,u_{cr}}^{t_i} \Big/ f_c^{t_i} \end{cases} \tag{10}$$

$$\sigma_s = \left[f_c^{t_i} - \sqrt{\left(f_c^{t_i}\right)^2 - 2 \times M_f \times f_c^{t_i} / \left[b(h_0)^2\right]} \right] \Big/ \rho_{s,u_{cr}}^{t_i} \tag{11}$$

$$\rho_{s,u_{cr}}^{t_i} = \left(A_{Re}^c - A_{cr}^{t_i}\right) / b_c h_0 \tag{12}$$

where M_f is the flexural moment; γ' is a correction factor when the compressive range is approximately rectangular, which can be assumed to be 1.0 when $f_c^{t_i}$ is less than 60 MPa [50]; h_0 is the effective height of the cross section (i.e., $h_0 = h_c - (d_c + d_{Re}/2)$); and $\rho_{s,u_{cr}}^{t_i}$ is the reinforcement ratio of the uncorroded area (Eq. (12)).

Given the strain profile (Fig. 7b), the thickness of the cracked layer ($h_{cr}^{t_i}$) can be determined as follows:

$$h_{cr}^{t_i} = h_c - \frac{\varepsilon_c^{t_i} h_0}{\beta} - \frac{\varepsilon_{cr}^c}{\varepsilon_s^{t_i}} x_s \tag{13}$$

where β is a correction factor of the compression height when the compressive range is approximately rectangular, which can be assumed to be 0.8 when $f_c^{t_i}$ is less than 60 MPa [50]; x_s is the distance between the neutral axis of the rebar's cross section and $x_s = h_0 - \varepsilon_c^{t_i} h_0 / \beta$; $\varepsilon_s^{t_i}$ is the strain of rebar where $\varepsilon_s^{t_i} = \sigma_s^{t_i} / E_s$; and ε_{cr}^c is the threshold cracking strain for chloride diffusion (see Section 5.1).

5.2. Effect of corrosion on the yield strength of rebar

Du et al. [51] conducted a comprehensive experiment to study the effect of corrosion on the yielding strength of embedded rebar. They found that the yield strength of embedded rebar declines linearly with its corrosion area ratio [51]. The yield strength of corroded rebar in this study, denoted as $f_{y,cor}^{t_i}$, is determined as follows:

$$f_{y,cor}^{t_i} = f_y \left(1 - \alpha_y \frac{A_{cor}^{t_i}}{A_{Re}^c} \times 100\right) \tag{14}$$

where f_y is the yield strength of non-corroded rebar; α_y is an empirical coefficient of the influence of corrosion on yield strength, which can be expressed as a beta distribution [51]; and A_{Re}^c is the area of the cross section of a rebar with a diameter of d_{Re}.

6. RC beam properties for analysis

This study simplifies the reliability analysis system by considering an RC beam subjected to flexural moment, as illustrated in Fig. 8. The RC beam has a cross section of $b_c \times h_c$ and a cover depth of d_c. Three rebars are laid on the bottom of the beam and are subjected to a flexural load. The properties of the RC beam are described in Sections 5.1 to 5.3.

6.1. Parameters for chloride diffusion

The chloride diffusion coefficient of uncracked concrete and the surface chloride content of concrete are given by Eqs. (2) and (3), respectively. The JSCE concrete design code [5] specifies a chloride diffusion coefficient for cracked concrete, which considers the effect of crack width through rebar stress (Eq. (16)). This study adopts this model for the chloride diffusion coefficient of cracked concrete. The effect of crack length is considered in the mechanical analysis (Eq. (13)). Page et al. [52] reported that the chloride diffusion coefficient of concrete depends on temperature, and this effect can be assessed using the Arrhenius equation. The temperature-dependence of chloride diffusion is determined by Eq. (17).

$$D_{con}^{t_i} = D_{con}^{t_i,T_0} m_{T_0}^{T_a} \tag{15}$$

$$D_{cr}^{t_i} = \left[D_{con}^{t_i,T_0} + 3(\sigma_s / E_s + \varepsilon_{sh}^{t_i}) D_{cr,k}\right] m_{T_0}^{T_a} \tag{16}$$

$$m_{T_0}^{T_a} = \exp\left[\frac{E_D}{R}\left(\frac{1}{T_0} - \frac{1}{T_a^{t_i}}\right)\right] \tag{17}$$

where E_s is the elastic module of the rebar; $\varepsilon_{sh}^{t_i}$ is the drying shrinkage strain of the concrete; $m_{T_0}^{T_a}$ is the temperature-dependent aspect of the chloride diffusion coefficient as calculated by the Arrhenius equation; E_D is the thermal activation energy of the chloride diffusion, which is taken as 4.0×10^4 J/mol [52]; and $D_{cr,k}$ is the characteristic chloride diffusion coefficient of the cracked concrete (54.8 mm^2/day) [5].

According to the JSCE concrete design code, the drying shrinkage of concrete ($\varepsilon_{sh}^{t_i}$) can be determined by Eq. (18) [5]:

$$\varepsilon_{sh}^{t_i} = \left\{1 - \exp\left[-0.108(t_i - t_0)^{0.56}\right]\right\} \varepsilon_{sh}^{u} \tag{18}$$

$$\varepsilon_{sh}^{u} = \gamma_{sh} \left\{-50 + 78[1 - \exp(RH/100)] + 38\ln W - 5[\ln(V/10S)]^2\right\} \times 10^{-5} \tag{19}$$

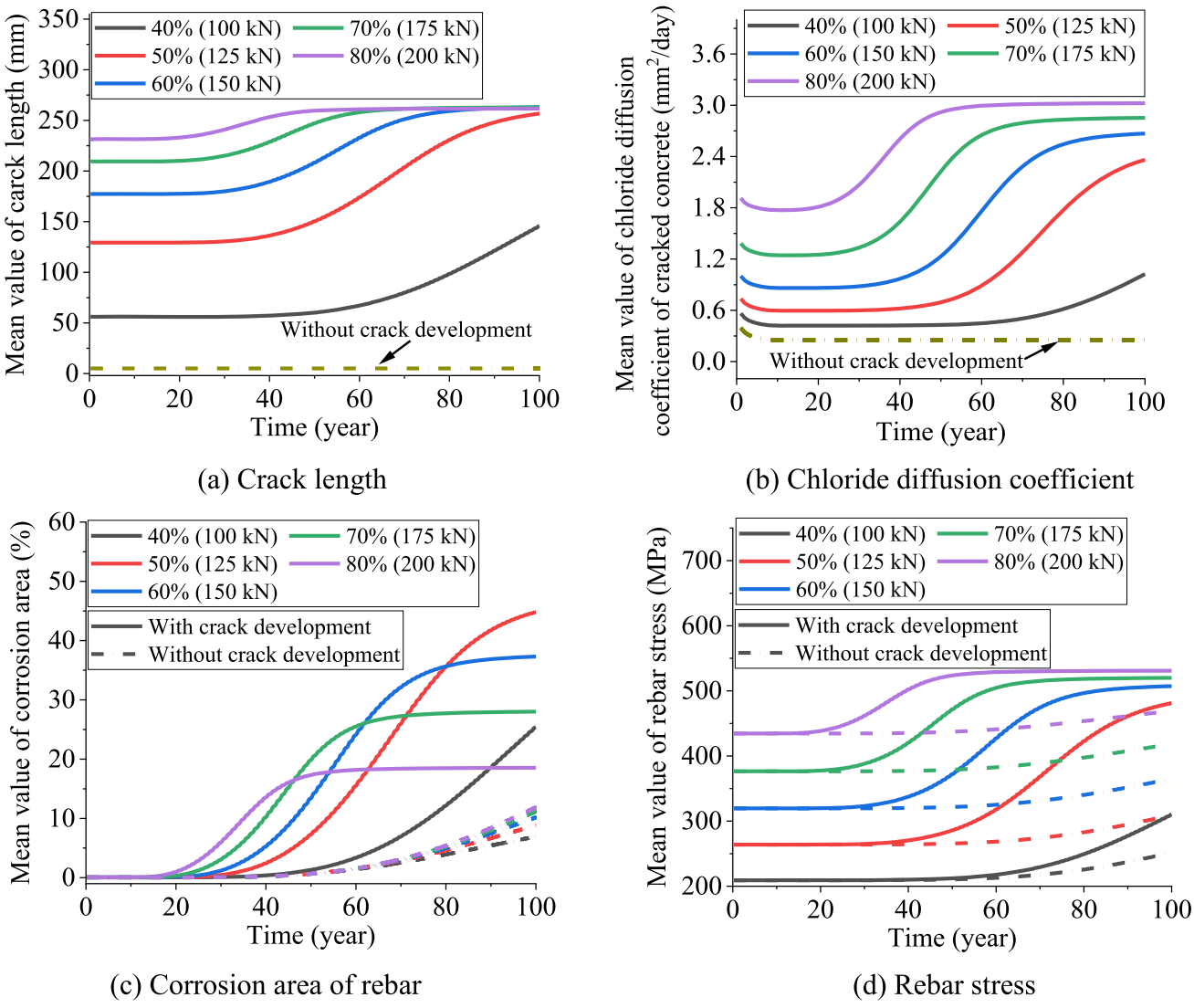


Fig. 9. Deterioration processes of reinforced concrete beam.

where γ_{sh} is a correction factor, which can be taken as 1.5; ε_{sh}^{u} is the ultimate drying shrinkage of the concrete (Eq. (19)); RH is the environmental humidity (%); W is the amount of water used for casting concrete (kg/m^3); and V/S is the volume-to-surface-area ratio of the RC beam (mm).

Djerbi et al. [53] studied the influence of crack width using the steady migration test, and reported that the influence can be ignored when the crack strain is less than 0.75×10^{-3}. It is noted that Eq. (15) considers the crack effect induced by drying shrinkage and its ultimate value is $\sim 0.45 \times 10^{-3}$. Therefore, for simplicity, the threshold cracking strain for chloride diffusion (ε_{cr}^{c}) is set as 1.2×10^{-3}.

6.2. Chloride binding effect

The chloride diffusion coefficients calculated in Eq. (15) represent the total chloride content profile in a concrete specimen. However, hydration products physically and chemically absorb chloride, such that only the free chloride in a pore solution contributes to corrosion [28,34]. The relationship between bound chloride content (C_{Re}^{b,t_i}) and free chloride content ($C_{\mathrm{Re}}^{free,t_i}$) can be expressed by the Langmuir isotherm [28,34]. Ishida et al. [28,41] developed a chloride binding model for cementitious material (Eq. (20)), and this study adopts their model, which is given as follows, to measure the binding effect:

$$C_{\mathrm{Re}}^{b,t_i} = \frac{\alpha_{cl}^{b} C_{\mathrm{Re}}^{free,t_i}}{1 + 4.0 \times C_{\mathrm{Re}}^{free,t_i}} \tag{20}$$

where α_{cl}^{b} is the chloride binding coefficient.

As the total chloride content is the sum of the bound chloride content and free chloride content, $C_{\mathrm{Re}}^{free,t_i}$ is given by Eq. (22).

$$b_{cl} = \alpha_{cl}^{b} + 1 - 4 \times C_{\mathrm{Re}}^{total,t_i} \tag{21}$$

$$C_{\mathrm{Re}}^{free,t_i} = \left(- b_{cl} + \sqrt{b_{cl}^2 + 16 \times C_{\mathrm{Re}}^{total,t_i}} \right) \Big/ 8 \tag{22}$$

where b_{cl} is an intermediate variable to determine the free chloride content of the concrete (Eq. (21)).

6.3. Compressive strength and pH

Both compressive strength ($f_c^{t_i}$) and pH (pH^{t_i}) are affected by cement hydration and the pozzolanic reaction of SCMs. For simplicity, in this study their time-dependencies are described using exponential functions, as follows:

$$f_c^{t_i} = f_c^{t_0} + \left(f_c^{u} - f_c^{t_0}\right)\left\{1 - \exp\left[\alpha_{Hy}\left(t_{Hy} + t_0\right)\right]\right\} \tag{23}$$

$$pH^{t_i} = pH^{t_0} + (pH^u - pH^{t_0})\{1 - \exp[\alpha_{Hy}(t_{Hy} + t_0)]\} \quad (24)$$

$$\begin{cases} t_{Hy} = t_i & t_i \le t^u_{Hy} \\ t_{Hy} = t^u_{Hy} & t_i > t^u_{Hy} \end{cases} \quad (25)$$

where $f_c^{t_0}$ and pH^{t_0} are the value of compressive strength and pH at reference age t_0, respectively; f_c^u and pH^u are the ultimate value of compressive strength and pH, respectively; α_{Hy} is the hydration rate factor that controls the compressive strength and pH development; t_{Hy} is the time variable for calculating pH and compression strength; and t^u_{Hy} is the time to hydration completion.

7. Results and discussion

The CFPs of an RC beam under various conditions are assessed in this section. The flexural capacity of each RC beam in the simulation is ~250 kN. The number of characteristic roots, interval time and interval depth of the chloride profile are 10, 0.25 year and 1 mm, respectively. The analysis presented in Section 2.2 confirms that these parameters can be used to provide a reasonable estimate of the chloride profile. The environmental temperature is the quarterly average temperature in Tokyo (Japan) from 1980 to 2020 (Fig. 15), as recorded by local weather bureaus [54]. Other parameters used in the analysis are given in Appendix E (Tables E1–E4).

It is assumed that the RC beam fails when the rebar stress ($\sigma_s^{t_i}$) is larger than then the yield strength of the corroded rebar ($f_{y,cor}^{t_i}$). Thus, the CFP of the RC structure at time $P_f(t_i)$ is given by Eq. (26). The sample size for the Monte Carlo simulation is 0.5 million. The convergence analysis in Appendix D shows that this sample size is large enough to provide a satisfactory estimate (Fig. D1). These analyses were conducted using parallel computing on a common computer (CPU Intel Xeon E5-2620, 2.10 GHz).

$$P_f(t_i) = \Pr\left(\sigma_s^{t_i} \ge f_{y,cor}^{t_i}\right) \quad (26)$$

7.1. Effect of external load level

The degradation processes of RC beam subjected to different flexural loads, including crack length, chloride diffusion coefficient of cracked concrete, corrosion area and rebar stress, are presented as the mean value for all Monte Carlo samples, as shown in Fig. 9. It also presents the CFP of RC without crack development, where the thickness of the cracked layer is set as a constant value of 5 mm (Fig. 9a). It should be noted the spatial variations of rebar corrosion is ignored in this study for simplification. It probably leads to an underestimation of the CFP of RC beams because the critical pitting corrosion is not always at midspan [12,55]. This effect should be carefully examined in the future.

Fig. 10 shows the effect of load level on the CFP of RC beams. Analysis results show that the rebar corrosion takes place after around 20 ~ 30 years (Fig. 9c), which is consistent with the results reported by F. Pugliese et al. [1]. The crack length (Fig. 9a) and chloride diffusion coefficient (Fig. 9b) then increase with the rebar corrosion (Fig. 9c). The rebar stress thus rapidly increases (Fig. 9d) until failure under the coupled degradation effects of crack development and rebar corrosion. Because the coupled degradation effects of crack development and rebar corrosion are ignored (Fig. 9), the CFP of an RC beam is significantly underestimated without crack development (Fig. 10). Specifically, the CFP of an RC beam is 93.82% after 100 years under 60% flexural capacity (150 kN) with crack development, whereas the corresponding CFP is only 10.20% without crack development (Fig. 10).

The results (Fig. 10) show that the CFP increases as the external load increases. That is because the crack length (Fig. 9a) and chloride diffusion coefficient (Fig. 9b) increase under higher external load at the beginning, which promotes the coupled degradation effects of crack development and rebar corrosion. On the other hand, the corrosion allowance of rebar decreases as the external load increases (Fig. 9c). Therefore, RC beam fails earlier under higher loads. Given a target CFP of 10%, the time at which maintenance of an RC beam is required decreases from 93.12 years to 33.72 years as the load increases from 40% to 80% flexural capacity (from 100 kN to 200 kN). These predictions agree well with the experimental observations reported by R. François et al. [29]. They monitored deterioration process of RC beams subjected to flexural load in a confined salt flog over 12 years.

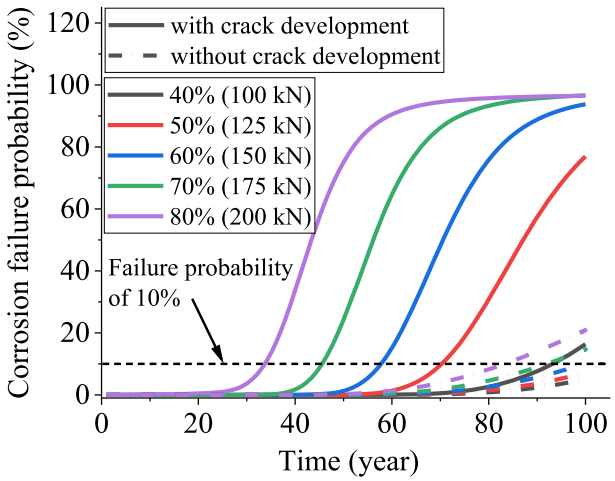


Fig. 10. Effect of loading on corrosion failure probability.

Table 3
Computational times in the proposed analysis model and the finite element method (FEM).

	40% (100 kN)	50% (125 kN)	60% (150 kN)	70% (175 kN)	80% (200 kN)
RNA-based analysis	114.52 h	99.51 h	84.47 h	66.31 h	50.83 h
FEM-based analysis	352.97 year	306.71 year	260.35 year	204.38 year	156.67 year

Table 3 summarizes the computational times required for analyses shown in Fig. 10. The computation effort based on FEM is estimated by assuming RNA to be 27,000 times faster than FEM (Table 2). Compared with the FEM-based method that takes hundreds of years for analysis, the RNA-based reliability analysis system developed in this study can be completed in 50 to 115 h on a common computer (Intel Xeon E5–2620, 2.10 GHz). This high-efficiency is extremely important for engineering practices, especially for estimating and optimizing whole-life cost of RC structure from the beginning.

7.2. Effect of SCMs

SCMs are widely used to improve the chloride resistance of RC structures. Fig. 11 compares the CFPs of OPC and SCM RC beams (i.e., fly ash, slag and LC3). The CFP of RC beams under 60% flexural capacity (150 kN) after 100 years is 93.82% for OPC, 59.99% for fly ash, 12.49% for LC3 and 7.18% for slag. These findings show that SCMs significantly improve the durability of RCs under chloride ingress and that slag provides better durability than fly ash, which are consistent with other experimental results [56]. Our analyses also show that the performance of LC3 RC is similar to that of slag RC (Fig. 11c); this is an important finding, as LC3 concrete is a promising green cementitious material.

Compared to OPC concrete, SCM concretes have a lower chloride diffusion coefficient (Fig. 12a), a lower pH of pore solution (Fig. 14a) and a higher chloride binding capacity (Fig. 13a). To determine the dominating factor in the optimization of SCMs in engineering practices,

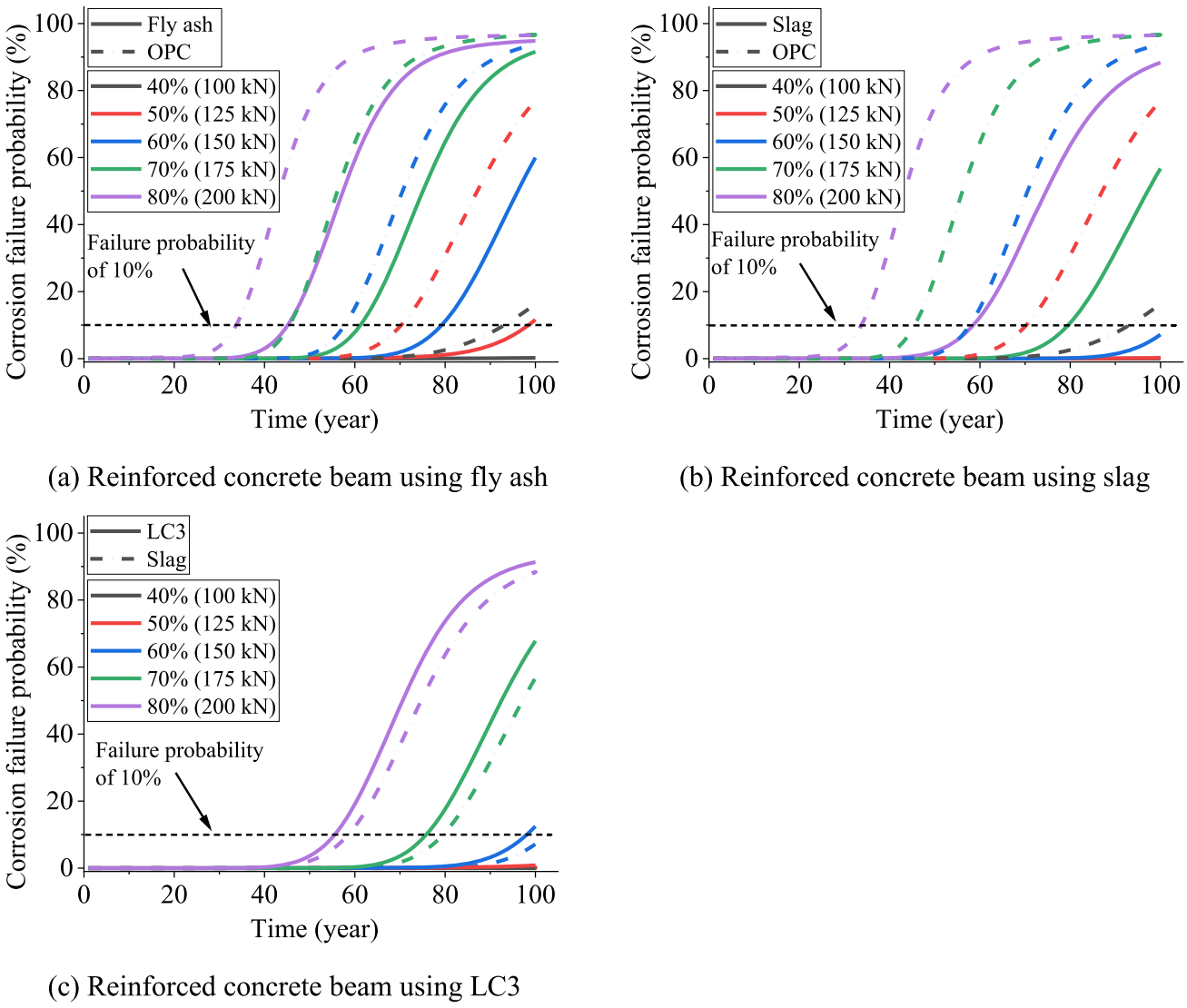


Fig. 11. Corrosion failure probability of reinforced concrete beam using SCMs.

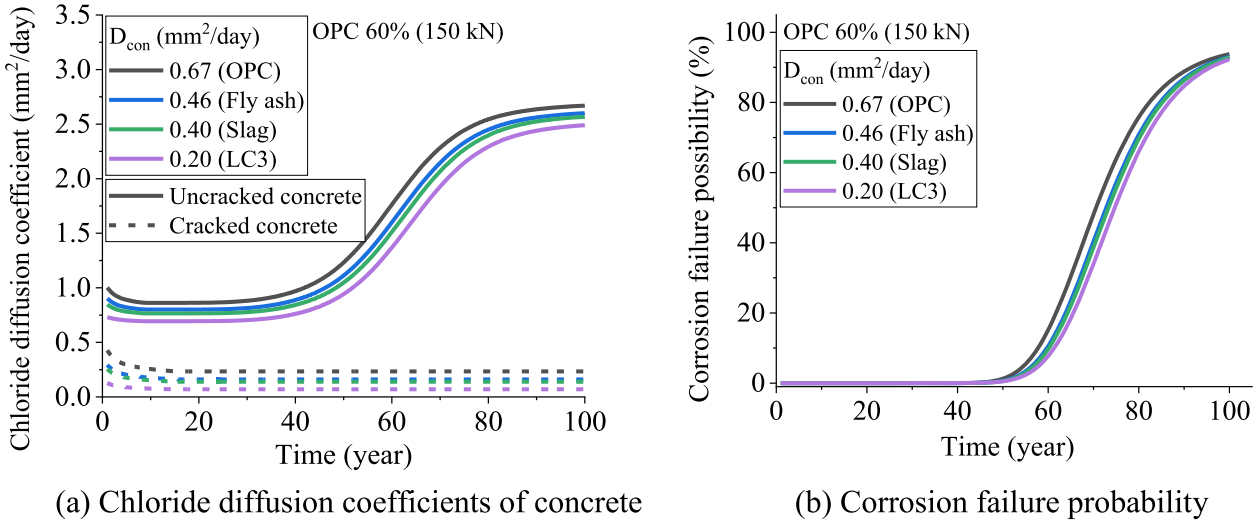


Fig. 12. Effect of chloride diffusion of concrete on corrosion failure probability for different SCMs.

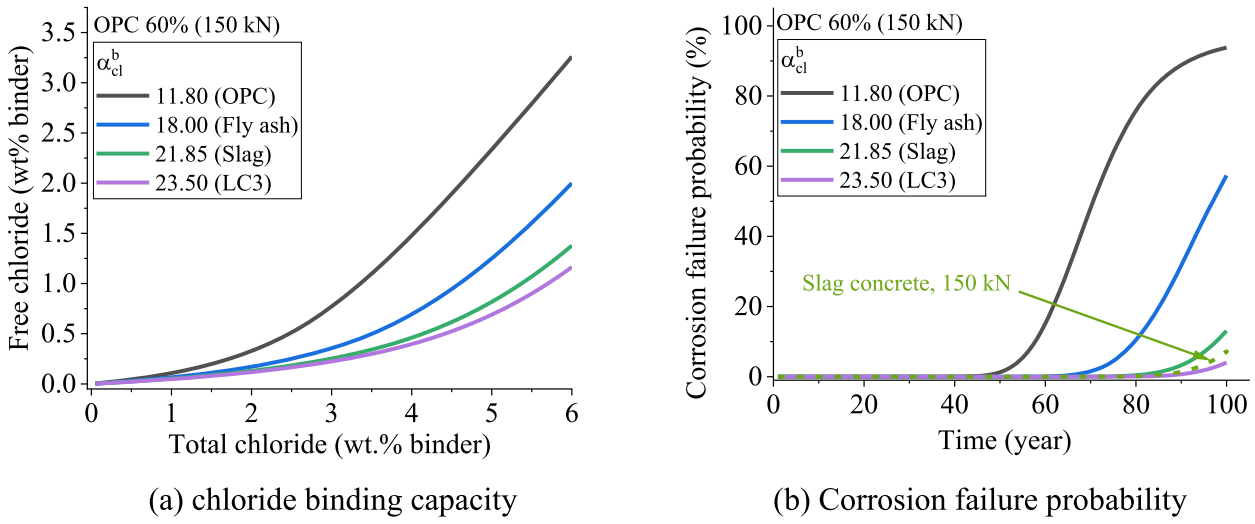


Fig. 13. Effect of chloride binding capacity on corrosion failure probability for different SCMs.

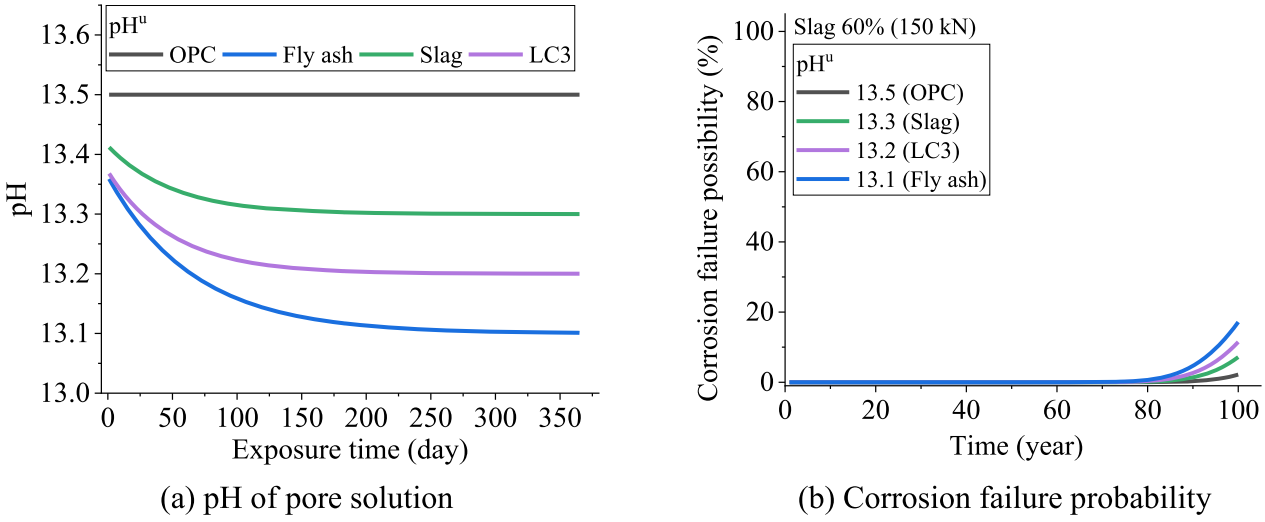


Fig. 14. Effect of pH on corrosion failure probability for different SCMs.

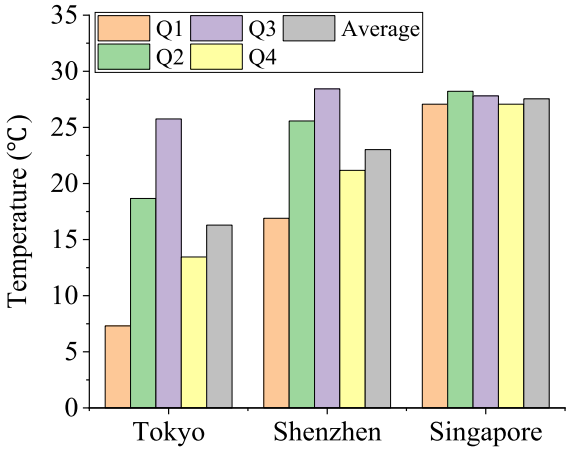
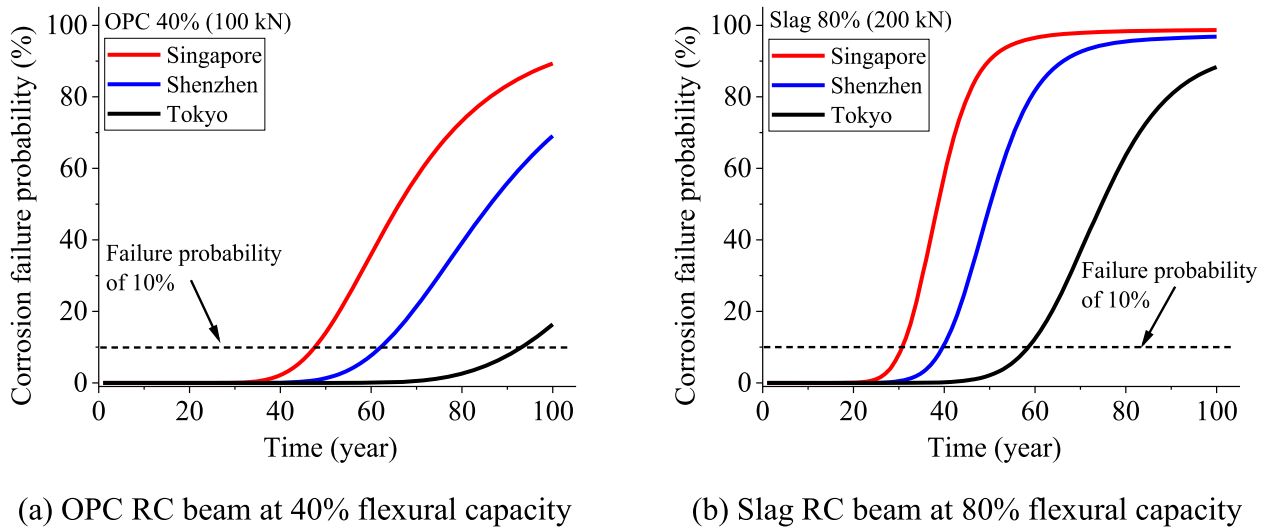


Fig. 15. Quarterly average temperatures of the three studied cities.

parameter analyses were conducted on an OPC RC beam and a slag RC beam under 60% flexural capacity (150 kN). Because the statistical information on chloride binding capacity and pH of SCMs blended cement paste is rarely documented, these two factors are deterministic in this study.

Fig. 12~ Fig. 14 show the results of the parameter analyses. The CFP increases as the chloride diffusion coefficient of concrete increases but decreases as the chloride binding capacity and pH of the pore solution increases. The effect of the chloride diffusion coefficient of concrete is not significant (Fig. 12b) because most chloride penetration occurs via cracks (Fig. 12a). In contrast, the chloride binding capacity has a significant effect on the CFP (Fig. 13): the CFP decreases from 93.82% to 13.08% as the chloride binding capacity (α_{cl}^{b}) increases from 11.80 to 23.50. The CFP of the OPC RC beam is close to that of the slag RC beam when they have the same chloride binding capacity (i.e., $\alpha_{cl}^{b} = 21.85$). Fig. 14 shows the effect of pH on the CFP of a slag RC beam under 60% flexural capacity (150 kN). The CFP increases from 2.18% to 17.15% as pH^{u} decreases from 13.5 to 13.1 over 100 years.

In conclusion, the chloride binding capacity is found to be the dominant parameter. As the incorporation of Al element significantly improves the chloride binding capacity of cement paste [34], the highly reactive aluminum-rich SCMs, such as (slag and LC3) are more suitable for RC structures under marine environments.



(a) OPC RC beam at 40% flexural capacity (b) Slag RC beam at 80% flexural capacity

Fig. 16. Parameter analyses of effects of environmental temperature on reinforced concrete beams in three cities.

7.3. Effect of environmental temperature

Many studies have reported that high temperatures promote both the diffusion of chloride in concrete and the corrosion rate of rebar [28,57–59]. The reliability analysis developed in this study considers the temperature-dependence of both chloride diffusion and corrosion rate, i.e., Eqs. (4), (15) and (16).

Temperature data obtained from three cities in different climate zones – Tokyo (Japan, from 1980 to 2020), Shenzhen (China, from 1981 to 2010) and Singapore (from 1981 to 2010) – are compared in this study. Fig. 15 shows the quarterly average temperatures of these three cities, based on data provided by local meteorological bureaus [54,60,61]. The average hottest city is Singapore and the average coolest city is Tokyo (Fig. 15).

Fig. 16 compares the CFPs of RC beams in these three cities. The degradation rate of an RC beam significantly increases at high temperatures. Given the target CFP of 10%, for an OPC RC beam under 40% flexural capacity (100 kN), the time at which maintenance is first required is 92.12 years in Tokyo, 61.13 years in Shenzhen and 46.58 years in Singapore. For a slag RC beam under 80% flexural capacity (200 kN), the time at which maintenance is first required is also significantly affected by temperature: it is 57.55 years in Tokyo, 38.69 years in Shenzhen and 29.74 years in Singapore. Therefore, temperature must be taken into account in the assessment of degradation of RC structures, especially as global temperatures keep on increasing with climate change.

8. Conclusions

This study develops a reliability analysis system that incorporates consideration of the coupled degradation effects of crack development and corrosion progression into assessment of the corrosion failure probability (CFP) of RC structures. As this system is developed based on physicochemical and mechanical models, it can be used to assess the whole-life performance of RC structures using OPC and supplementary cementing materials (SCMs) without the need for empirical deterioration models for RC. Moreover, this system incorporates an RNA of chloride diffusion and is thus computationally efficient, such that whole-life assessment of RC structures can be completed within dozens of hours on a desktop computer. Therefore, it is a practical system for assessing the CFP of RC structures subjected to various loads and environmental attacks.

The results of analyses show that RC deterioration processes are significantly underestimated if the coupled effects of crack development and corrosion progression on RC are not considered. In addition, the results show that the durability of RC structures can be significantly improved by using SCMs. Although the chloride diffusion coefficient of SCM-blended concrete is lower than that of OPC, this does not significantly affect the structural integrity of cracked RC structures because most chloride penetration occurs via cracks, so that chloride binding capacity is the dominant parameter. Highly reactive aluminum-rich SCMs, such as slag and LC3, are thus more suitable than other SCMs for use in RC structures in marine environments, as the incorporation of Al significantly improves the chloride binding capacity of cement paste. The effect of environmental temperature is also studied by comparing the CFPs of RC beams in three cities. The results show that the CFP of RC beams significantly increases as temperature increases. This system could provide engineers with useful information to incorporate sufficient safety factors to address the increasing temperature due to climate change.

However, this system is a prototype and has several limitations. First, the cement hydration and pozzolanic reaction model of SCMs is not considered. Accordingly, a hydration model and corresponding microstructure model should be explored in future research, as these models enable accurate determination of RC properties such as compressive strength, pH and chloride diffusion coefficients. Second, a cross-section analysis is used in the flexural analysis to develop the reliability system. However, a three-dimensional mechanical analysis model should also be explored in future research to account for the spatial variability associated with steel corrosion in longitudinal rebars. Third, only the flexural load is addressed in this study is examined, so future research should examine other types of mechanical degradation, such as shear failure and corrosion expansion. Fourth, several types of experiments should be conducted to verify the reliability analysis system, such as salt-water ponding tests of cracked concrete, chloride binding tests, and fatigue bending tests of corroded RC beams. Moreover, cyclic loading conditions, such as traffic loading and earthquake loading, could also be considered in future work to increase the applicability of these models to various loading scenarios.

CRediT authorship contribution statement

Tiao Wang: Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing, Project administration, Funding acquisition, Resources. **Chunhe Li:** Methodology, Writing – review & editing, Resources. **Jian-jun Zheng:** Methodology, Writing – review & editing. **Jürgen Hackl:** Methodology, Writing – review & editing. **Yao Luan:** Methodology, Writing – review & editing. **Tetsuya**

Ishida: Methodology, Writing – review & editing. **Satya Medepalli:** Formal analysis, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Analytical solution of chloride diffusion in a two-layered material

An analytical solution of chloride diffusion in a two-layered material has been developed. Fig. A1 shows the chloride diffusion model, in which the thickness and the chloride diffusion coefficient are h_1 and D_1 for the first layer, and h_2 and D_2 for the second layer, respectively.

The chloride profile of two-layered materials can be determined as

$$C(x,t) = C^t_{x=0} + \sum_{n=1}^{\infty} A_n f_n(x,\lambda_n)\exp\left(-\lambda_n^2 t\right) \tag{A1}$$

where $C^t_{x=0}$ is the surface chloride content; x is the depth; t is the exposure time; and $f_n(x,\lambda_n)$ is defined as

$$f_n(x,\lambda_n) = \begin{cases} \dfrac{\cot\left(\lambda_n h_2 \Big/ \sqrt{D_2}\right)}{\sin\left(\lambda_n h_1 \Big/ \sqrt{D_1}\right)} \sin\left(\lambda_n x \Big/ \sqrt{D_1}\right) & 0 \le x \le h_1 \\ \dfrac{1}{\sin\left(\lambda_n h_2 \Big/ \sqrt{D_2}\right)} \cos\left[\lambda_n (x - h_1 - h_2) \Big/ \sqrt{D_2}\right] & h_1 \le x < h_1 + h_2 \end{cases} \tag{A2}$$

A_n represents the influence of existing chloride content, which is determined by Eq. (A3):

$$A_n = \int_0^{h_1+h_2} \left[\overline{C}(x) - C^t_{x=0}\right] f_n(x,\lambda_n)dx \Big/ \int_0^{h_1+h_2} f_n^2(x,\lambda_n)dx \tag{A3}$$

where $\overline{C}(x)$ is the chloride profile in the last step; and

$$\begin{aligned} \int_0^{h_1+h_2} f_n^2(x,\lambda_n)dx \quad &= \frac{\cot^2\left(\lambda_n h_2 \Big/ \sqrt{D_2}\right)}{2\sin^2\left(\lambda_n h_1 \Big/ \sqrt{D_1}\right)} \left[h_1 - \frac{\sqrt{D_1}}{2\lambda_n}\sin\left(\frac{2\lambda_n h_1}{\sqrt{D_1}}\right)\right] \\ &+ \frac{1}{2\sin^2\left(\lambda_n h_2 \Big/ \sqrt{D_2}\right)} \left[h_2 + \frac{\sqrt{D_2}}{2\lambda_n}\sin\left(\frac{2\lambda_n h_2}{\sqrt{D_2}}\right)\right] \end{aligned} \tag{A4}$$

The characteristic roots (λ_n) can be determined using Eq. (A5) which searches each semi-period region of $\tan(\lambda_n h_1/\sqrt{D_1})$ (or $\tan(\lambda_n h_2/\sqrt{D_2})$) using the binary search method. Algorithm details can be found in reference [11].

$$\tan\left(\lambda_n h_1 \Big/ \sqrt{D_1}\right)\tan\left(\lambda_n h_2 \Big/ \sqrt{D_2}\right) = \sqrt{D_1/D_2} \tag{A5}$$

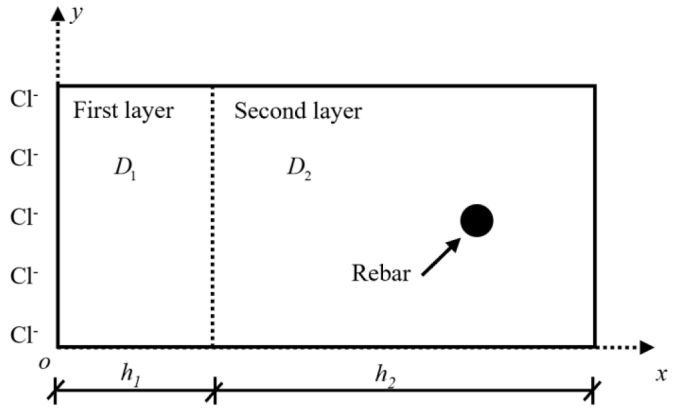


Fig. A1. Chloride diffusion model for a two-layered material.

The influence of the last-step chloride content is considered by parameter A_n^i. If the chloride profile $\overline{C^{i-1}}(x)$ at the *i-1th* step is expressed in terms of discrete points, A_n^i at the ith step can be obtained by substituting $\overline{C^{i-1}}(x)$ into Eq. (A3) and then integrating it into each point, as follows

$$A_n^i \approx \sum_l^L \left[\overline{C^{i-1}}(x_l) - C_{x=0}^i\right] f_n^i(x_l, \lambda_n)\Delta x \Big/ \int_0^{h_1+h_2} \left(f_n^i(x, \lambda_n)\right)^2 dx \tag{A6}$$

where x_l is the specimen depth at the lth point, L is the total number of points, $f_n^i(x_l, \lambda_n)$ is the trigonometric function at the ith step, and Δx is the interval.

Appendix B. Corrosion model based on an Evans' diagram

Corrosion is the result of two half-cell reactions, an anodic reaction (oxidation) and a cathodic reaction (reduction), that results from an ionic current flowing through a concrete pore [62,63]. The logarithmic current is linearly proportional to the corrosion potential, which is illustrated in a Tafel plot (Fig. B1) [63]. This relationship describes both the reduction and oxidation reactions. As corrosion occurs when the oxygen reduction reaction and the reinforcement oxidation reaction have the same reaction rate, the corrosion current and the corrosion potential is the point where the Tafel plots of the reduction reaction and oxidation reaction meet, which is depicted in an Evans' diagram (Fig. B1) [64].

Recent research has indicated that the corrosion current also depends on the hydroxide-to-chloride ratio of the pore solution [38]. Hussain et al. [28,40] have developed a corrosion current model based on an Evans' diagram, in which the influence of pH and chloride ions is assessed by adjusting the potential of the oxygen reduction reaction (E_{O_2}) and the potential of the iron oxidation reaction (E_{Fe}) at the reference corrosion current (i.e., I_{Fe}^{Θ} and $I_{O_2}^{\Theta}$). The corrosion current thus increases as chloride content increases and pH decreases, as shown in Fig. B1. The following modified model is developed, based on the model in Hussain et al. [28,40], to determine the corrosion current I_{cor} of rebar:

$$\log(I_{cor}) = \frac{\log\left(I_{Fe}^{\Theta}\right) \times \eta_{Fe} - \log\left(I_{O_2}^{\Theta}\right) \times \eta_{O_2} + E_{O_2} - E_{Fe}}{\eta_{Fe} - \eta_{O_2}} \tag{B1}$$

$$\eta_{O_2} = -\frac{2.303 \times RT_a}{0.5 \times Z_{O_2} F} f_{O_2} \tag{B2}$$

$$\eta_{Fe} = \frac{2.303 \times RT_a}{0.5 \times Z_{Fe} F} \tag{B3}$$

$$E_{O_2} = E_{O_2}^{\Theta} + \left(\frac{RT_a}{Z_{O_2} F}\right) \ln C_{O_2} - \chi_{pH} \times pH \tag{B4}$$

$$E_{Fe} = E_{Fe}^{\Theta} + \left(\frac{RT_a}{Z_{Fe} F}\right) \ln C_{Fe} \times P_{Cl} \tag{B5}$$

$$P_{Cl} = \chi_{cl} + 3 \times 10^{-3} \times \ln\left(1.0 + 1 \times 10^4 \cdot C_{cor}^{free}\right) + 1.0 \tag{B6}$$

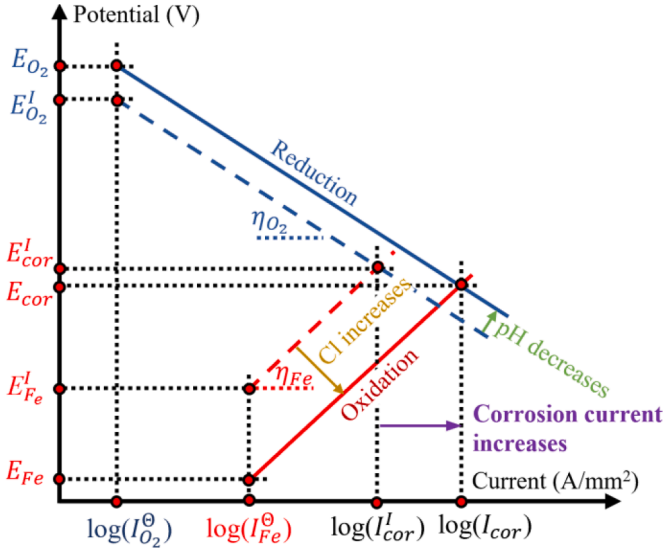


Fig. B1. Corrosion current model based on an Evans' diagram. $E_{O_2}^I$, E_{Fe}^I and E_{cor}^I are the potentials for oxygen reduction reactions and iron oxidation reactions, respectively, when the cement paste is common OPC paste without chloride ingress (i.e., pH = 13.5 and $C_{free}^{Re} = 0$) at T_0 and I_{cor}^I is the corresponding current. Other parameter definitions can be found in this section.

$$\chi_{Cl}=\begin{cases}\dfrac{2.0\times C_{cor}^{free}}{1.0+C_{cor}^{free}} & C_{cor}^{free}\le 0.4\\ 0.571+\dfrac{0.175\times\left(C_{cor}^{free}-0.4\right)}{1.0+\left(C_{cor}^{free}-0.4\right)} & 0.4<C_{cor}^{free}\le 2.0\\ 0.679+\dfrac{0.165\times\left(C_{cor}^{free}-2.0\right)}{1.0+\left(C_{cor}^{free}-2.0\right)} & C_{cor}^{free}>2.0\end{cases} \tag{B7}$$

$$C_{cor}^{free}=C_{\mathrm{Re}}^{free}+C_{cement} \tag{B8}$$

where I_{Fe}^{Θ} and $I_{O_2}^{\Theta}$ are the reference corrosion current at the anode and cathode, adjusted for corrosion potentials, and are 1.0×10^{-5} and 1.0×10^{-10} A/m^2, respectively; η_{Fe} and η_{O_2} are the Tafel slopes of the reduction and the oxidation reactions, respectively; Z_{Fe} and Z_{O_2} are the number of valence electrons of iron and oxygen, which are both 2; F is Faraday's constant, which is 9.65×10^4 C/mol; f_{O_2} is a correction factor for the Tafel slope of the reduction reaction, which is assumed to be 2.0 in this study; C_{cor}^{free} is the free chloride content (wt% binder), which is determined by Eq. (B8); C_{cement} is the free chloride content of the raw cement and SCMs, which is assumed to be 2.25×10^{-2} (wt% binder); E_{O_2} and E_{Fe} are the corrosion potential of the oxygen reduction reaction and the iron oxidation reaction, respectively (V); $E_{O_2}^{\Theta}$ and E_{Fe}^{Θ} are the reference corrosion potential of the oxygen reduction reaction and the iron oxidation D_{Re}^{D} reaction, which are assumed to be 1.226 V and −0.44 V (referenced to a standard hydrogen electrode), respectively; C_{O_2} and C_{Fe} are the concentrations of dissolved oxygen and iron in the pore solution, which for simplification are assumed to be 0.2 μmol/L and 1.0×10^{-6} μmol/L, respectively; pH is the pH value of the pore solution; P_{Cl} is the adjusting factor for the corrosion potential of the iron oxidation reaction under the influence of the free chloride in the pore solution, which is determined by Eq. (B6); χ_{pH} is a factor measuring the influence of pH, which is taken as 0.06 [28]; and χ_{Cl} is a factor measuring the influence of the free chloride in the pore solution, which is determined by Eq. (B7).

Appendix C. Numerical approach to determining the area of pitting corrosion

This section provides an efficient approach to determining the area of a rebar affected by pitting corrosion. A cross-section of the rebar is divided into rectangles with edges of d_e (Eq. (C1)), as shown in Fig. C1. The cross-section of rebar A_{Re}^{c} can thus be approximately determined using Eq. (C2)

$$d_e=d_{Re}\,/\,D_{Re}^{D} \tag{C1}$$

$$A_{\mathrm{Re}}^{c}=M_T d_e^2 \tag{C2}$$

where D_{Re}^{D} is the divided number for the rebar diameter; and M_T is the number of total rectangles.

The distance from a rectangle of interest (x_I, y_I) to the bottom rectangle (x_B, y_B) is given by Eq. (C3)

$$P_d^I=\sqrt{(x_B-x_I)^2+(y_B-y_I)^2}\,d_e \tag{C3}$$

The corroded rectangle can be efficiently determined by Eq. (C4):

$$\begin{cases}Corroded & P_d^I\le P_{cor,pit}\\ Uncorroded & P_d^I>P_{cor,pit}\end{cases} \tag{C4}$$

The pitting corrosion area is thus given by Eq. (C5):

$$A_{cor}=M_{cor}d_e^2 \tag{C5}$$

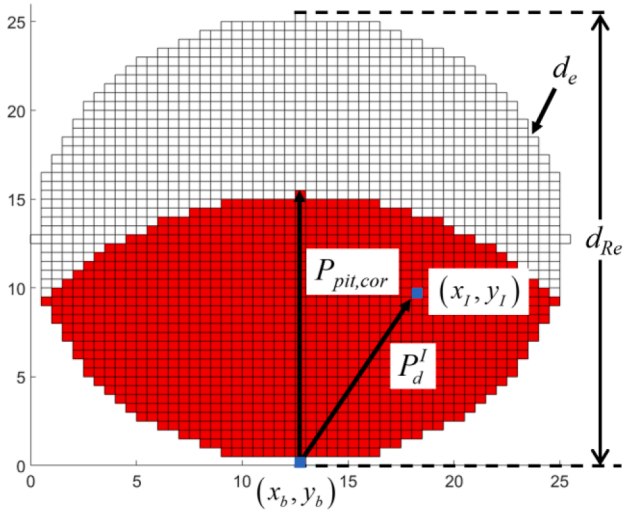


Fig. C1. Numerical approach for determining the area of the corroded region.

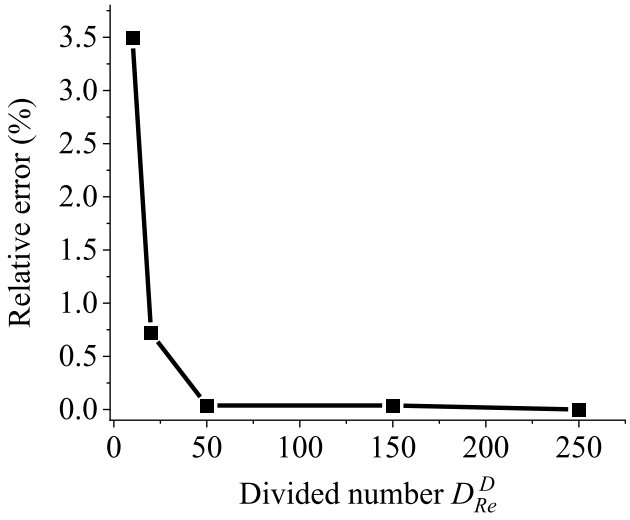


Fig. C2. Convergent analyses of the divided number (D_{Re}^{D}).

where M_{cor} is the number of corroded rectangles.

A_{Re}^{c} and A_{cor} approach the correct values as D_{Re}^{D} increases. A relative error, Err_{cor}, defined by Eq. (C6), is used to estimate the effect of D_{Re}^{D}. The diameter of the rebar (d_{Re}) and pitting corrosion depth ($P_{cor,pit}$) are assumed to be 25 mm and 12.5 mm, respectively. The corroded area A_{cor}^{250} is assumed to be 'correct' when D_{Re}^{D} is 250. Fig. C2 shows the convergent analyses using D_{Re}^{D}. The relative error is almost equal to 0 when D_{Re}^{D} is greater than 50. Therefore, is set as 50 in this study.

$$Err_{cor} = \sqrt{\left(A_{cor}^{D_{Re}^{D}} - A_{cor}^{250}\right)^{2}} \Big/ A_{cor}^{250} \tag{C6}$$

where $A_{cor}^{D_{Re}^{D}}$ is the corroded area when the divided number is D_{Re}^{D}.

Appendix D. Convergent analysis of sample size

This appendix presents the convergent analysis of the sample size. Fig. D1 compares the probability density distribution of the CFP of an OPC RC under a load of 175 kN for different sample sizes. A sample size of two million is assumed to be an 'accurate' result because it is sufficiently large. The results of the convergent analysis show that a sample size of 0.5 million provides a satisfactory estimate. Therefore, this study uses sample sizes of 0.5 million for every case.

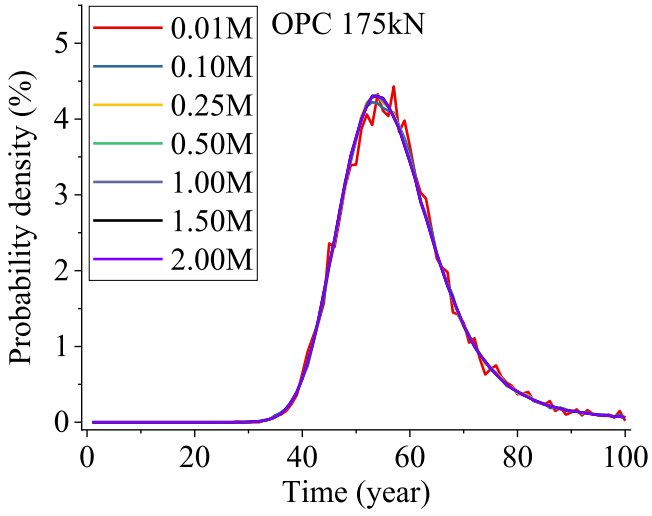


Fig. D1. convergent analysis of sample size (M represents million).

Appendix E. Parameters considered in the analyses[1]

Table E1
Reinforced concrete beam parameters.

Symbol	Description	Type	Mean	CoV	Unit	Ref.
L_s	Distance from edge to support	Constant	350		mm	
L_a	Distance from support to load	Constant	1000		mm	
L_f	Length between two loads	Constant	600		mm	
b_c	Beam width	Norm	250.0	0.02	mm	[25]
h_c	Beam height	Norm	450.0	0.02	mm	[25]
h_{cov}	Concrete cover depth	Norm	100.0	0.15	mm	[11]
d_{Re}	Diameter of flexural rebar	Norm	25.0	0.02	mm	[25]

Table E2
Reinforced concrete parameters.

Symbol	Description	Type	Mean	CoV	Unit	Ref.
wb	Water-binder ratio	Beta	0.45	0.025		
$D_{con}^{t_0}$	Chloride diffusion coefficient at age of t_0					
	OPC concrete	Log norm	0.67	0.20	mm^2/day	[11,65]
	Fly ash concrete	Log norm	0.46	0.20	mm^2/day	[11,65]
	Slag concrete	Log norm	0.40	0.20	mm^2/day	[11,66]
	LC3 concrete	Log norm	0.20	0.15	mm^2/day	[67]
α_{con}	Time-dependence of diffusion coefficient					
	OPC concrete	Beta	0.20	0.30		[65]
	Fly ash concrete	Beta	0.44	0.30		[65]
	Slag concrete	Beta	0.25	0.30		[66]
	LC3 concrete	Beta	0.50	0.25		[67]
$f_c^{t_0}$	Compression strength at age of t_0					
	OPC concrete	Log norm	15.0	0.10	MPa	[68]
	Fly ash concrete	Log norm	12.0	0.10	MPa	[68]
	Slag concrete	Log norm	13.5	0.10	MPa	[69]
	LC3 concrete	Log norm	15.0	0.10	MPa	[67]
f_c^u	Ultimate compression strength					
	OPC concrete	Log norm	55.0	0.10	MPa	[68]
	fly ash concrete	Log norm	60.0	0.10	MPa	[68]
	Slag concrete	Log norm	60.0	0.10	MPa	[69]
	LC3 concrete	Log norm	55.0	0.10	MPa	[67]
pH^{t_0}	pH value of cement paste at age t_0					
	OPC concrete	Constant	13.5			[35]
	Fly ash concrete	Constant	13.5			[35]
	Slag concrete	Constant	13.5			[35]
	LC3 concrete	Constant	13.5			[70]
pH^u	Ultimate pH value of cement paste					
	OPC concrete	Constant	13.5			[35]
	Fly ash concrete	Constant	13.1			[35]
	Slag concrete	Constant	13.3			[35]
	LC3 concrete	Constant	13.2			[70]
t_{Hy}^u	Time to hydration completion					
	OPC concrete	Constant	8.0		year	
	Fly ash concrete	Constant	15.0		year	
	Slag concrete	Constant	12.5		year	
	LC3 concrete	Constant	12.5		year	
α_{Hy}	Hydration factor of cement paste					
	OPC concrete	Constant	− 0.030			[71]
	Fly ash concrete	Constant	− 0.015			[71]
	Slag concrete	Constant	− 0.020			[72]
	LC3 concrete	Constant	− 0.020			[73]
α_{cl}^b	Chloride binding factor of cement paste					
	OPC concrete	Constant	11.80			[28,34]
	Fly ash concrete	Constant	18.00			[28,34]
	Slag concrete	Constant	21.85			[28,34]
	LC3 concrete	Constant	23.50			[34,74]
W	Water amount used for casting concrete	Constant	175.0		kg/m^3	

[1] 1. The variables without a reference are empirical values used in this study due to lack of suitable reference values.2. CoV = coefficient of variation; Ref. = reference.3. OPC = ordinary Portland cement; LC3 = limestone calcined clay cement.4. Constant = constant value; Norm = normal distribution; Log norm = lognormal distribution; Beta = beta distribution; and Gumbel = Gumbel distribution.

Table E3
Rebar parameters.

Symbol	Description	Type	Mean	CoV	Unit	Ref.
f_y	Yield strength of rebar	Norm	550.0	0.03	MPa	[25,75]
E_s	Elastic modulus of rebar	Norm	206.0	0.03	GPa	[25,75]
α_y	Corrosion factor for yield strength	Norm	0.002	0.01		[51,75]
P_{pit}	Pitting corrosion factor	Gumbel	8.324	0.157		[12]

Table E4
Environmental parameters.

Symbol	Description	Type	Value	Unit	Ref.
		Data source / Equation			
RH	Humidity	Constant	90	%	
$C^a_{x=0}$	Initial surface chloride content factor	Constant	$-0.379 \times wb + 2.064$	(wt% binder)	[43]
$C^b_{x=0}$	Time-dependent surface chloride content factor	Constant	$4.078 \times wb + 1.011$	(wt% binder)	[43]
T_{En}	Temperature	Data from local weather bureaus		°C	[54,60,61]

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